EE400D - Electrical Engineering Design Project

Professor: Dr. Saleh Al Jufout

Due Date: April 5, 2026

Group 1

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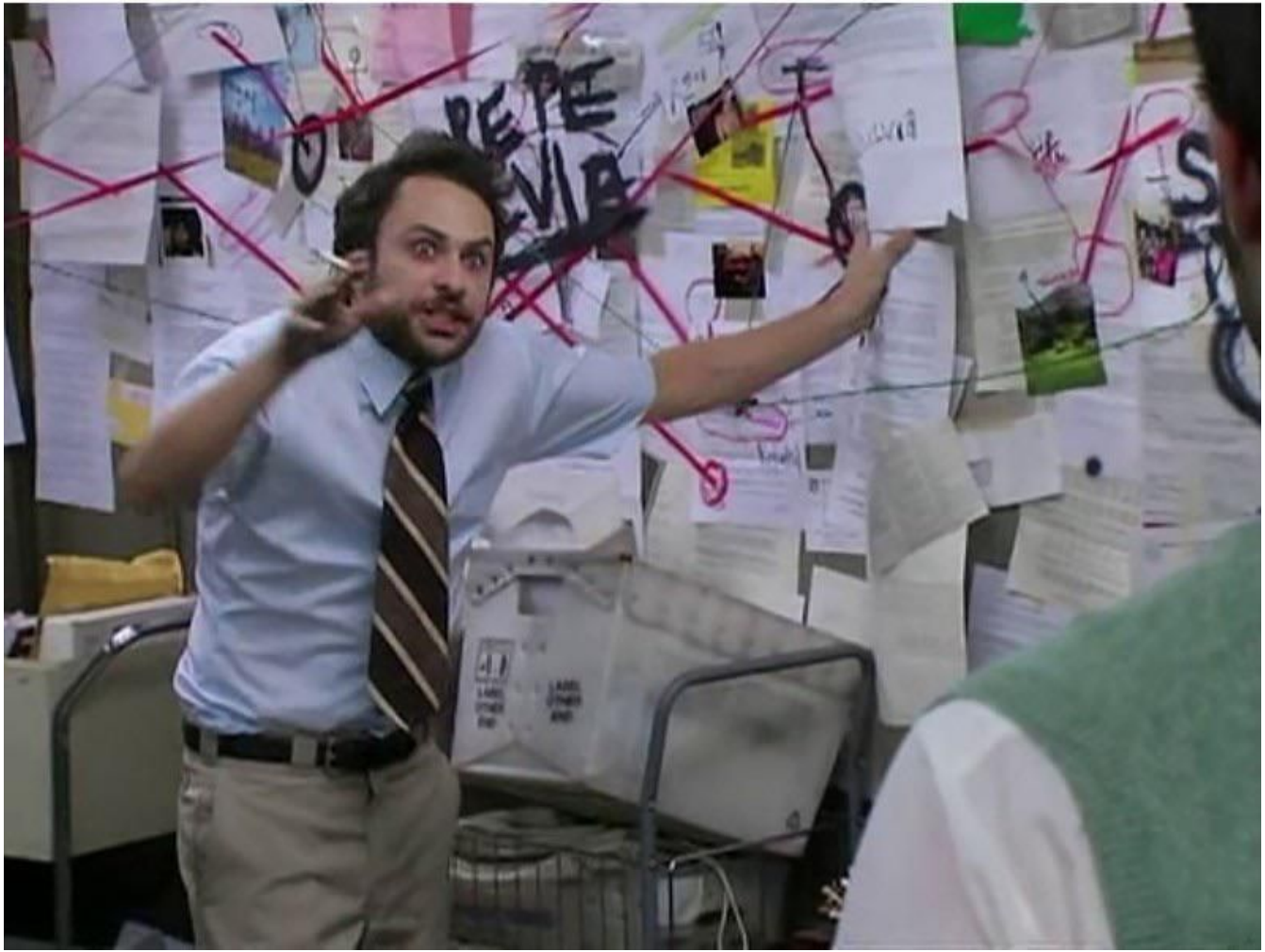
LONG BEACH STATE UNIVERSITY

College of Engineering



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Project Description / Overview

Define the Problem

- ESCs are designed for narrow, single-use applications
- Limited reuse due to fixed voltage and control designs
- Modular, universal ESC architecture proposed
- Modules are interchangeable and scalable
- Adaptable to drones, tools, and robotics
- Novel stackable hardware design

Problem Statement: *Current ESCs are application-specific and inflexible, creating a need for a modular, scalable, and universal motor control platform.*

Project Objectives

- **Design a modular ESC system** where identical motor-control modules can be added or removed to support different motor counts (ex: 2, 4, or 6 motors) without redesigning the entire controller.
- **Maintain consistent performance as the system scales**, including stable motor startup, throttle response, and speed control across all connected modules.
- **Provide a standardized interface** for power, control, and communication so the ESC integrates cleanly with common flight controllers or robotics controllers.
- **Implement core safety behaviors** to protect the platform, including overcurrent, undervoltage, and over temperature response (fault detection and safe shutdown).
- **Support real-time monitoring** so users can view key operating data per module (ex: current, voltage, temperature, RPM or commutation status) for debugging and system health.
- **Improve serviceability and maintainability** by enabling a failed module to be swapped without replacing the entire ESC system.
- **Demonstrate repeatable operation** under realistic loading conditions, showing the system can run continuously without instability, excessive heating, or unexpected resets.

2. Mission Profile / Demonstration

Demonstration Focus

The final demonstration will verify that the modular ESC system can scale to different motor counts, operate reliably under load, and maintain safe, stable control while modules are added or removed.

Success will be confirmed through a live bench demonstration using a configurable motor test setup, supported by measured electrical performance and system-level safety checks.

Demonstration Plan (What we will show)

We will assemble the ESC in two configurations (ex: 1-motor and 4-motor) by adding/removing identical power modules, then run each setup on a motor test stand with propellers removed (or a dynamometer if available).

The system must demonstrate: smooth startup, stable speed control, and consistent telemetry/feedback while staying within safe limits for temperature, current ripple, and fault response (overcurrent and undervoltage protections will be intentionally triggered and verified).

3. Constraints

- Project will not exceed $400.
- Project will be completed by May 6th, 2026.
- Must comply with basic electrical safety standards.
- Must operate within safe temperature limits.
- Lead times for parts and PCBs
- Modularity tradeoff: modular design adds cost/volume vs single-purpose ESCs.

4. Requirements

Level 1 Requirements:

- The ESC system shall drive BLDC motors.
- The system shall support modular stacking of identical ESC units.
- The total prototype system cost shall not exceed $400.
- The system shall operate safely within standard battery voltage ranges (16.8V).
- The system shall provide stable and smooth motor control under all rated operating conditions.
- The architecture shall be scalable and adaptable across multiple platforms (drones, tools, robotics).

Level 2 Requirements

- Input voltage range shall be 9V–30V DC.
- Maximum continuous current should be up to 30A.
- PWM switching frequency shall be configurable between 20–40 kHz.
- System efficiency should be ≥ 10% at nominal load.
- Overcurrent and overvoltage protection shall be implemented in hardware.
- Overtemperature shutdown shall activate at ≤120°C junction temperature.

Level 2 Mechanical Requirements

- Module shall support stacking of up to 4 identical ESC boards.
- Board-to-board connectors shall support up to 30A continuous current.
- PCB copper thickness shall be able to support up to 30A
- 3D-Printed enclosure to ensure no copper can be touched by user

Level 2 Electrical Requirements

- Commutation shall remain stable up to 40 kHz PWM frequency.
- Firmware shall include watchdog timer and fail-safe shutdown.
- Communication interface shall support CAN or UART protocol.

Verification Test Plan

Motor Drive and Control Verification

- Verify the ESC can start and drive a BLDC motor reliably
- Test low, medium, and high throttle commands
- Confirm smooth startup and stable speed control

Pass: Motor starts, runs continuously, and responds correctly to commands

Output Current and PWM Verification

- Verify operation across the supported output current range
- Test PWM operation at key switching frequencies
- Measure waveform quality and motor stability

Pass: ESC operates correctly without instability across tested conditions

Protection System Verification

- Verify overcurrent, undervoltage, and overtemperature protection
- Confirm proper warning, shutdown, or safe-state response

Pass: System protects hardware and transitions safely during fault events

Verification Test Plan Continued

Communication Verification

- Verify communication through serial comms protocol (CAN, UART, SPI, etc.)
- Confirm real-time reporting of voltage, current, temperature, and motor status
- Compare transmitted data to measured bench values

Pass: Commands are received correctly and telemetry is accurate

Modularity Verification

- Verify operation with stacked ESC modules
- Test at least a 2-board configuration, with expansion toward multi-motor operation
- Confirm stable operation and communication across modules

Pass: System operates correctly as identical modules are added

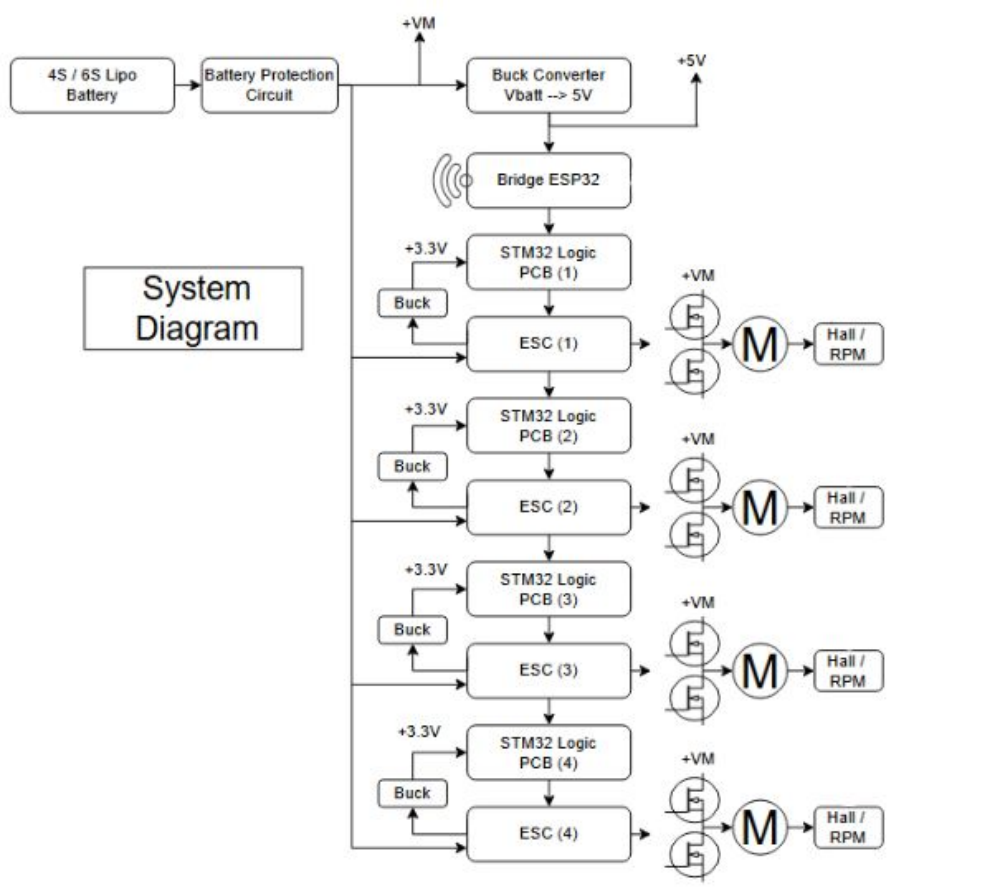
Verification Summary

- **Functional testing:** motor drive, startup, speed control
- **Electrical testing:** voltage range, PWM operation
- **Safety testing:** fault protection and shutdown response
- **System testing:** communication, telemetry, and modular expansion

Verification Test Matrix

Req. Number	Requirement	Test Case	Verification Method
1	ESC shall drive BLDC motors and provide smooth, stable motor control	Run one BLDC motor at low, medium, and high throttle; verify smooth startup and stable speed control	Functional Test
2	ESC shall operate safely over the supported input range and maintain stable PWM operation	Test operation across supported current range and verify PWM at key switching frequencies	Electrical Test
3	ESC shall implement core protection features	Introduce controlled overcurrent, undervoltage, and over temperature fault conditions and verify safe response	Fault/Safety Test
4	ESC shall support communication and telemetry	Send commands through serial communication and verify reporting of voltage, current, temperature, and motor status	Communication Test
5	ESC shall support modular stacking of identical ESC units	Test at least a 2-board configuration and verify stable operation, control, and communication as modules are added	System Test

System Block Diagram



1. 4S / 6S Lipo battery goes directly into a protection circuit with:

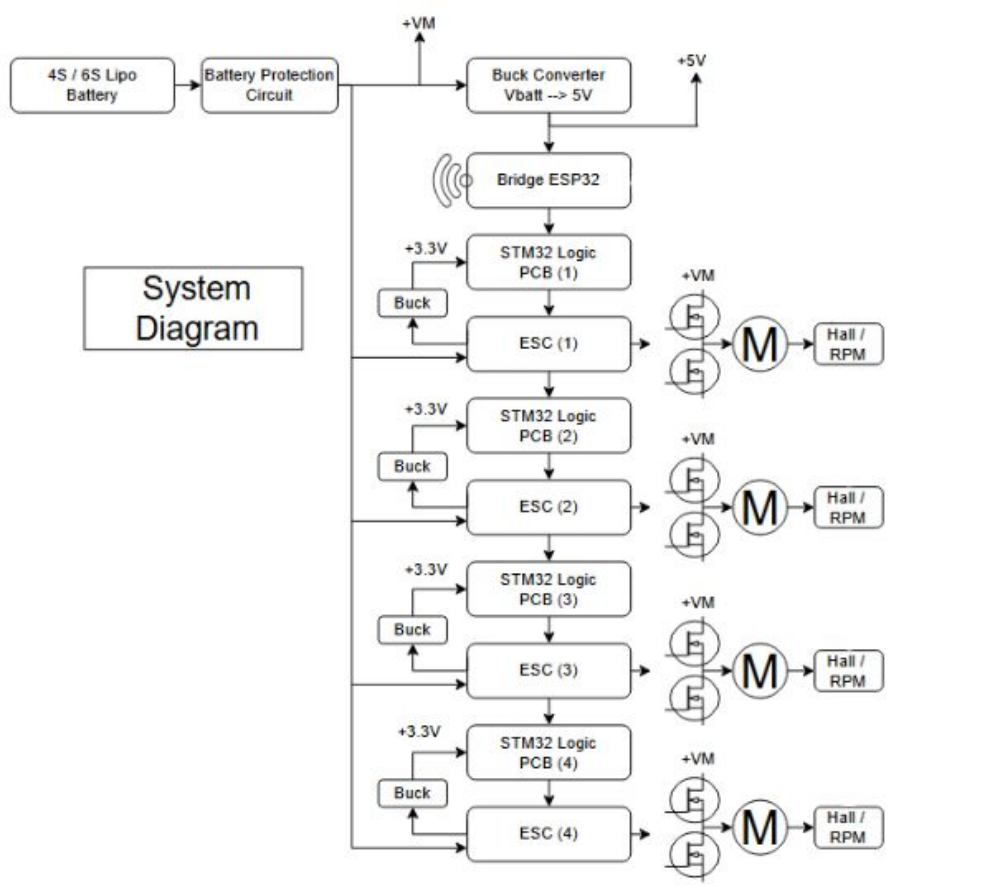
- Rev polarity, over voltage, & over current detection with automatic system shutdown

2. Protection circuit outputs to motor bus which provides power to motors and 5V/5A buck converter which supplies ESP32s

3. Motor Driver IC has a built in buck to power STM32s (3.3V)

4. ESP32s constantly relaying important information to each other

System Block Diagram

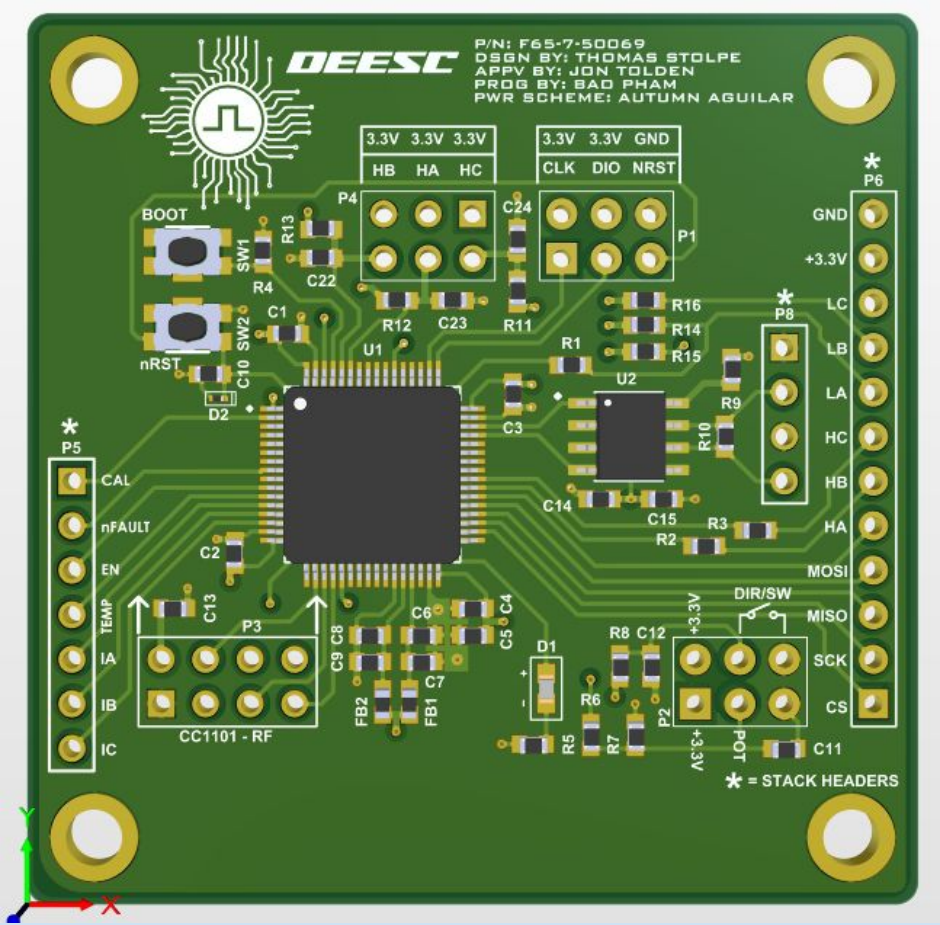


5. STM32 generates 3 phase PWM waveforms, which are amplified by the BLDC motor driver ICs

6. Motor Driver IC amplifies PWM through charge pump capacitors to ensure half bridge gates turn on. It also automatically reports problems to MCU

7. Hall effect sensors read analog value from motors magnetic field to measure RPM

Electrical Design MCU / Logic Control PCB



- 2x2 inch overall size
- **STM32G474RET6** MCU

a. 3 x 16-bit 8-channel advanced motor control timers

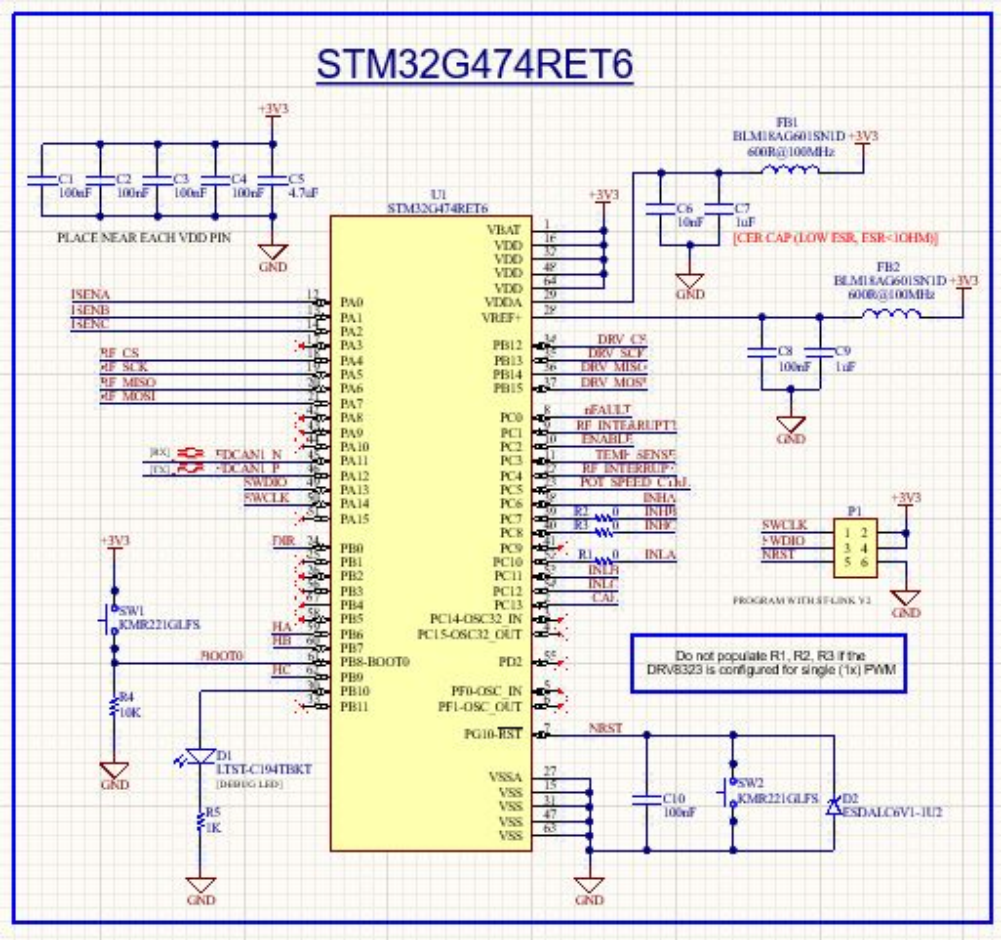
b. 3 x 16-bit 8-channel advanced motor control timers + deadtime

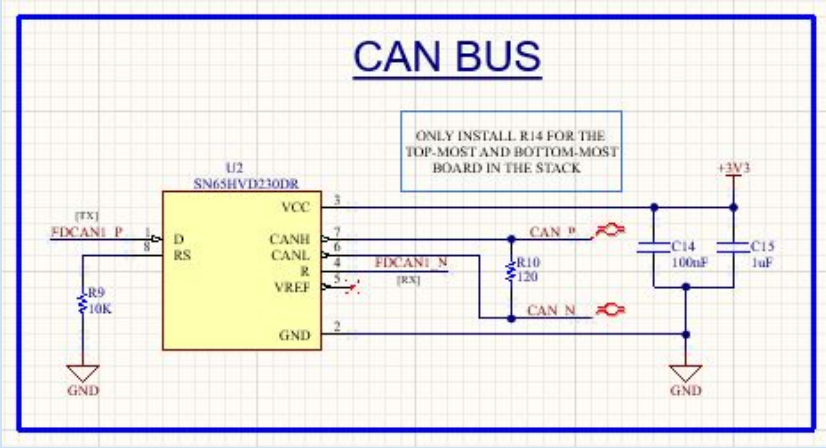
c. x5 12 bit ADC pins – we used all of them

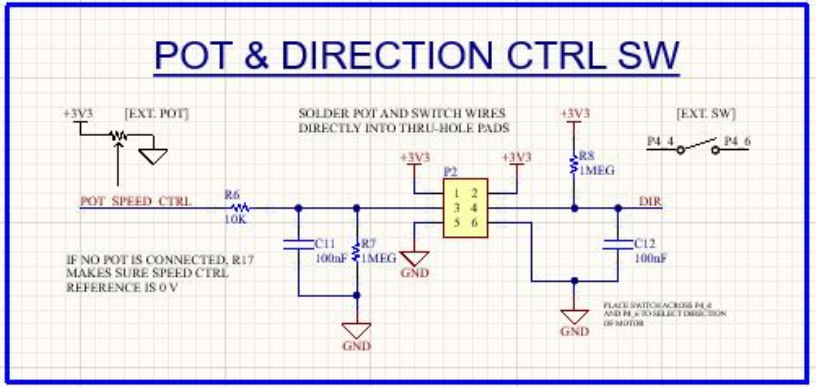
d. X3 CAN bus, 4 SPIs, x5 UARTs

e. 170MHz internal clock speed without external OSC crystal

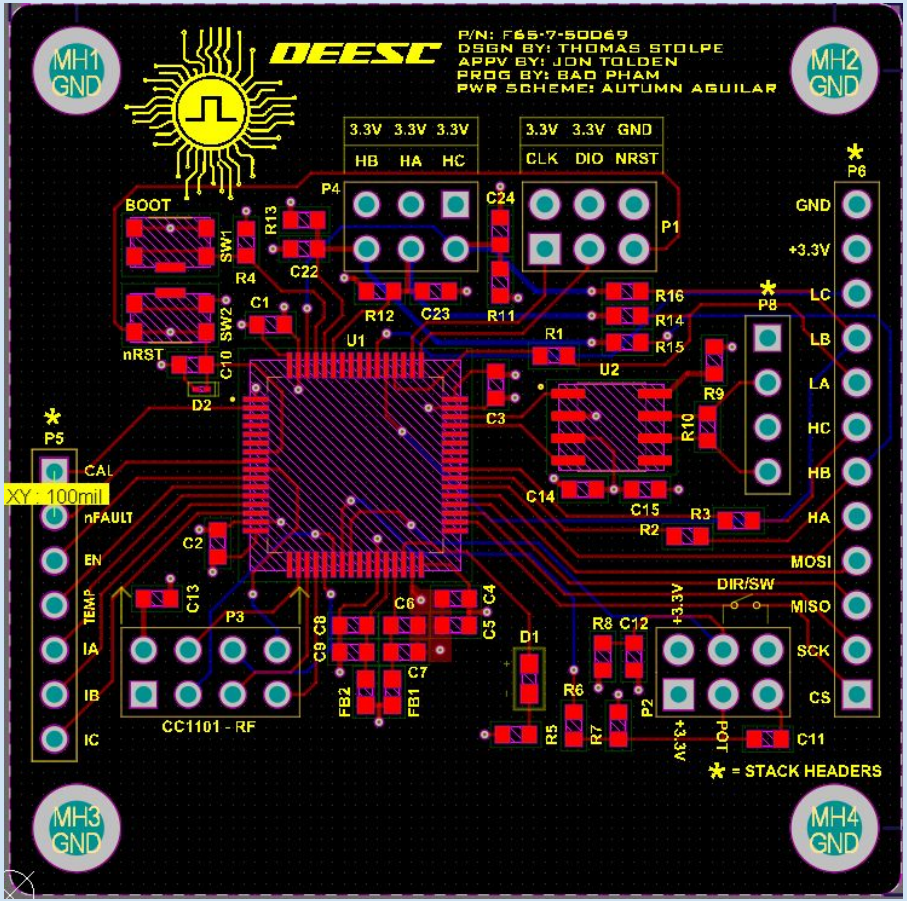
Electrical Schematic MCU / Logic Control PCB

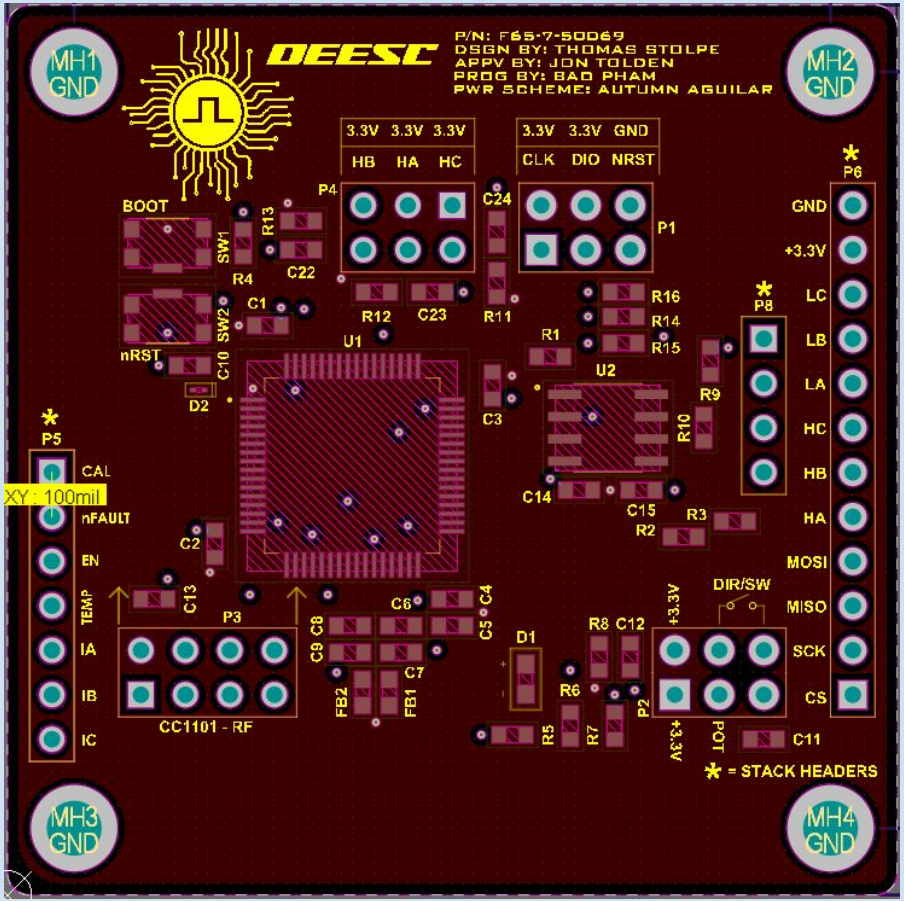




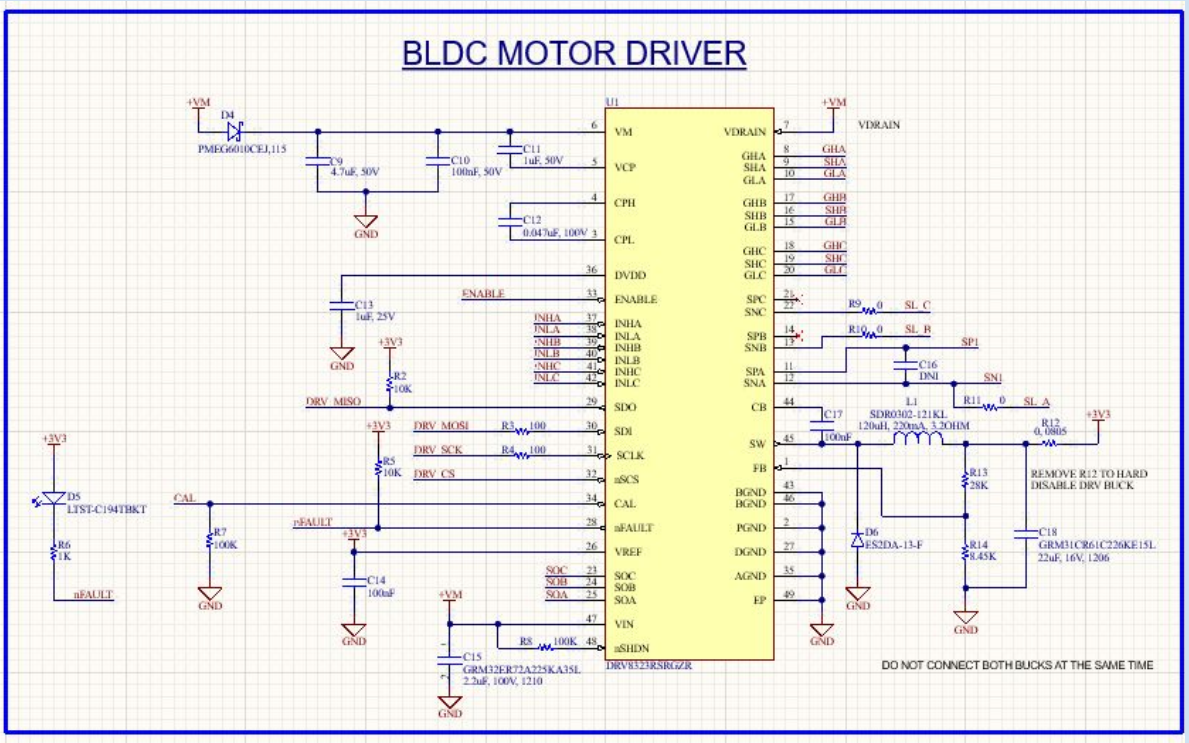


MCU PCB - Layout





ESC PCB



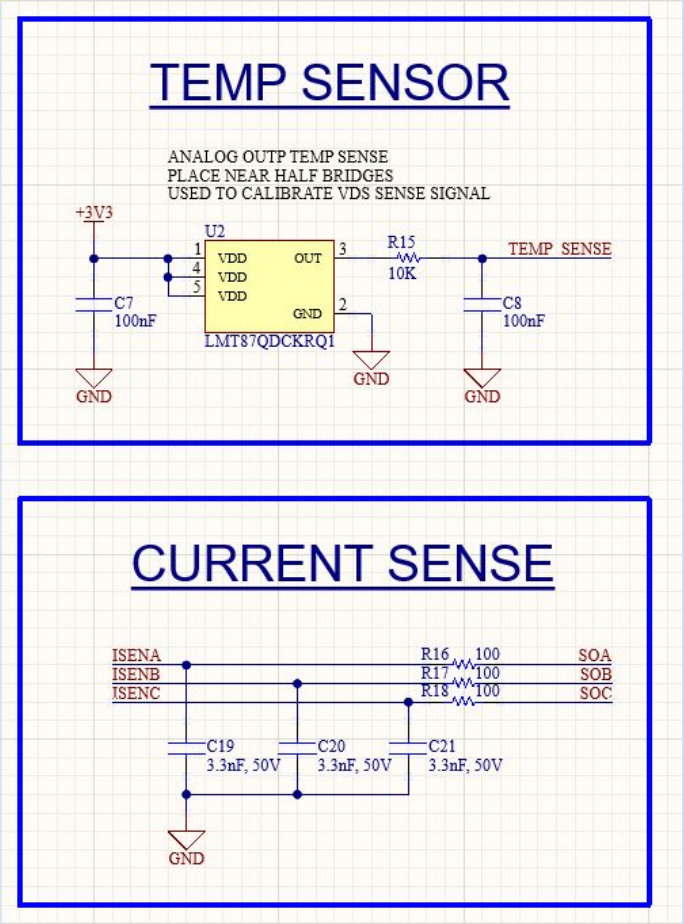
1. Integrated buck converter

2. SPI configurable for faults (over-volt, over-curr, over-temp, etc)

3. Built-in current shunt amplifiers – allows for feedback

4. Built in protection:
   a. Enable pin
   b. nFAULT
   c. Overshoot protection
   d. Deadtime
   e. Built in current sensing

ESC PCB - Peripherals



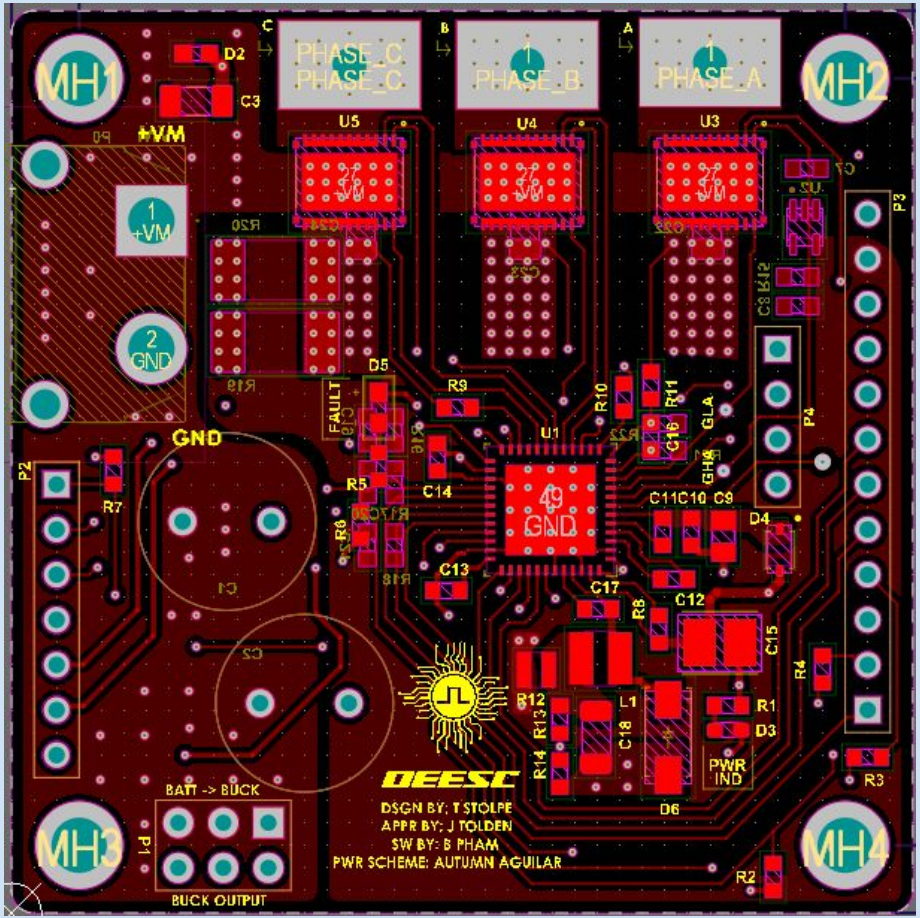
Temp Sensor

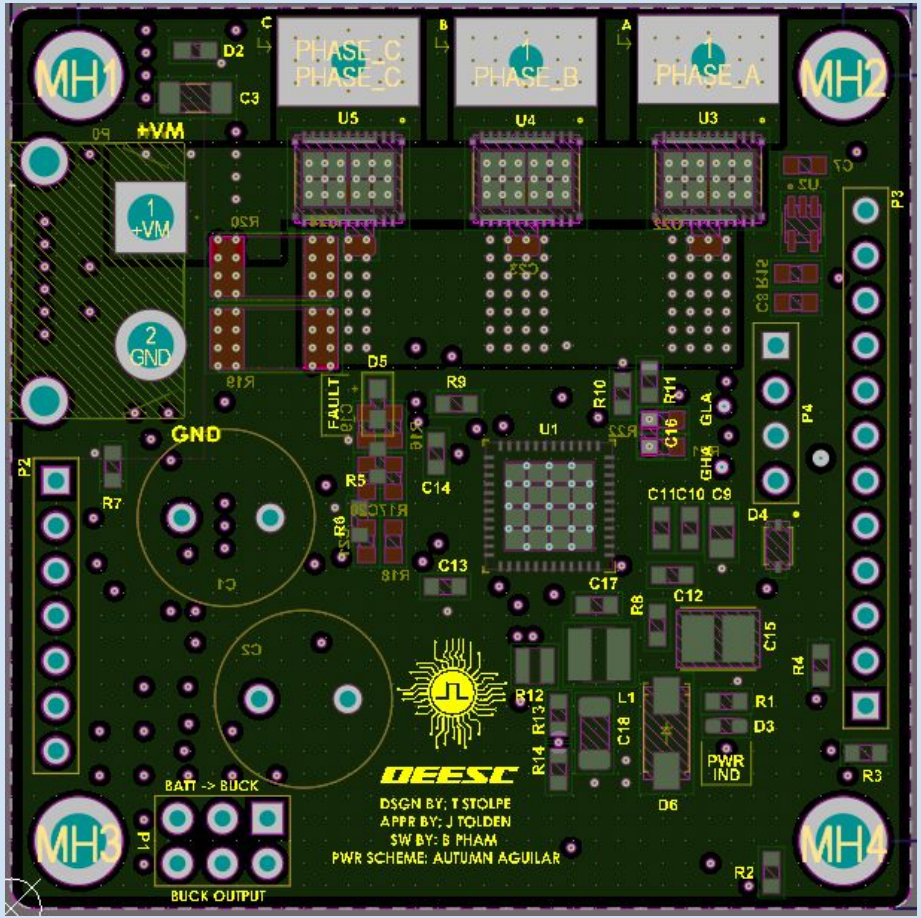
1. Takes temperature of circuit board directly above high current trace to motor
2. If temperature exceeds 150 deg C, motors automatically shutdown to prevent MOSFETs from getting damaged

Current Sense

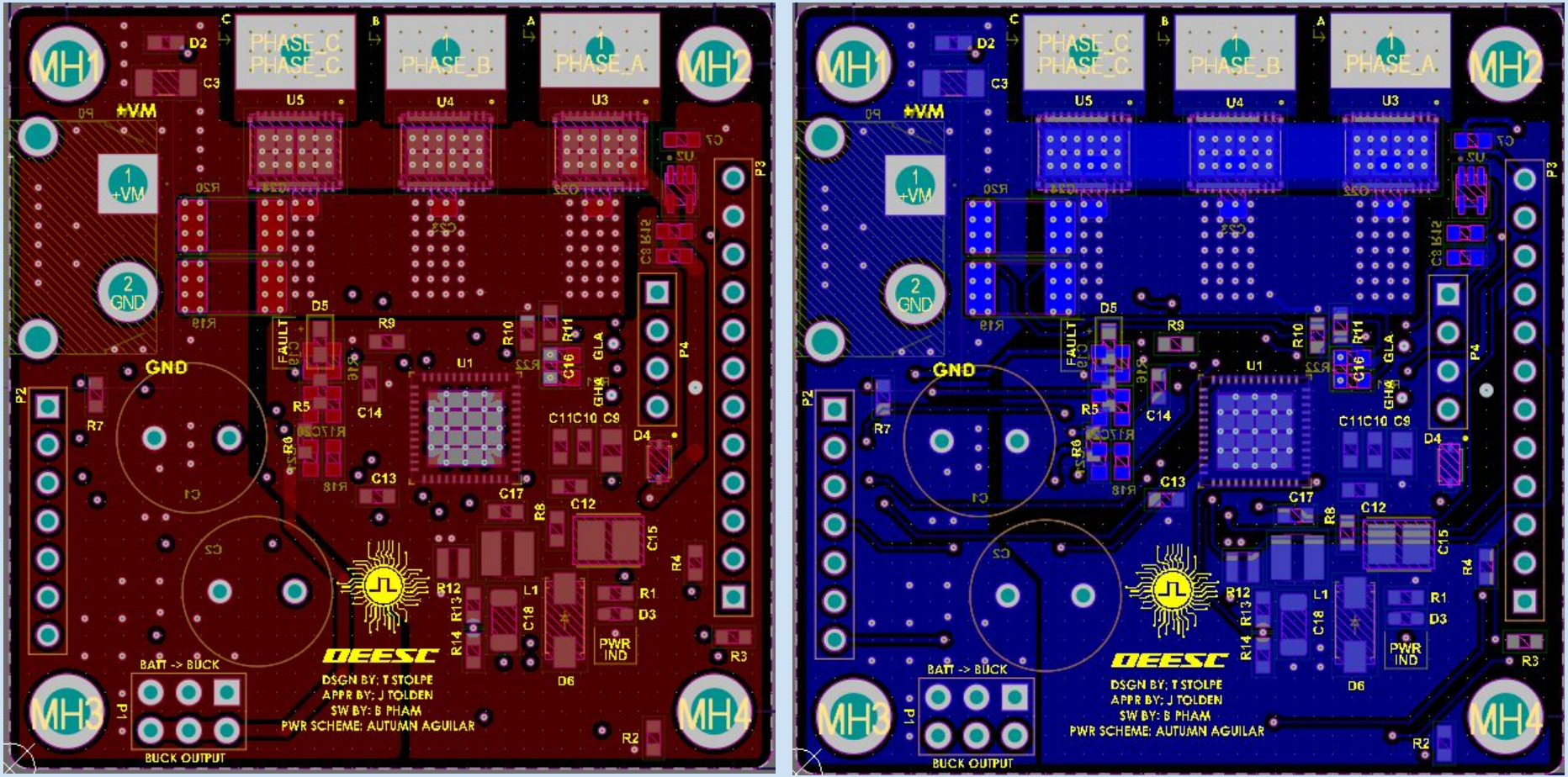
1. Current sense amplifier measures tiny voltage drop across shunt resistor and amplifies it with gain = 40
2. Allows accurate motor current measurement and protect circuit from overcurrent and allows for feedback control loops

ESC Board / PCB Layout

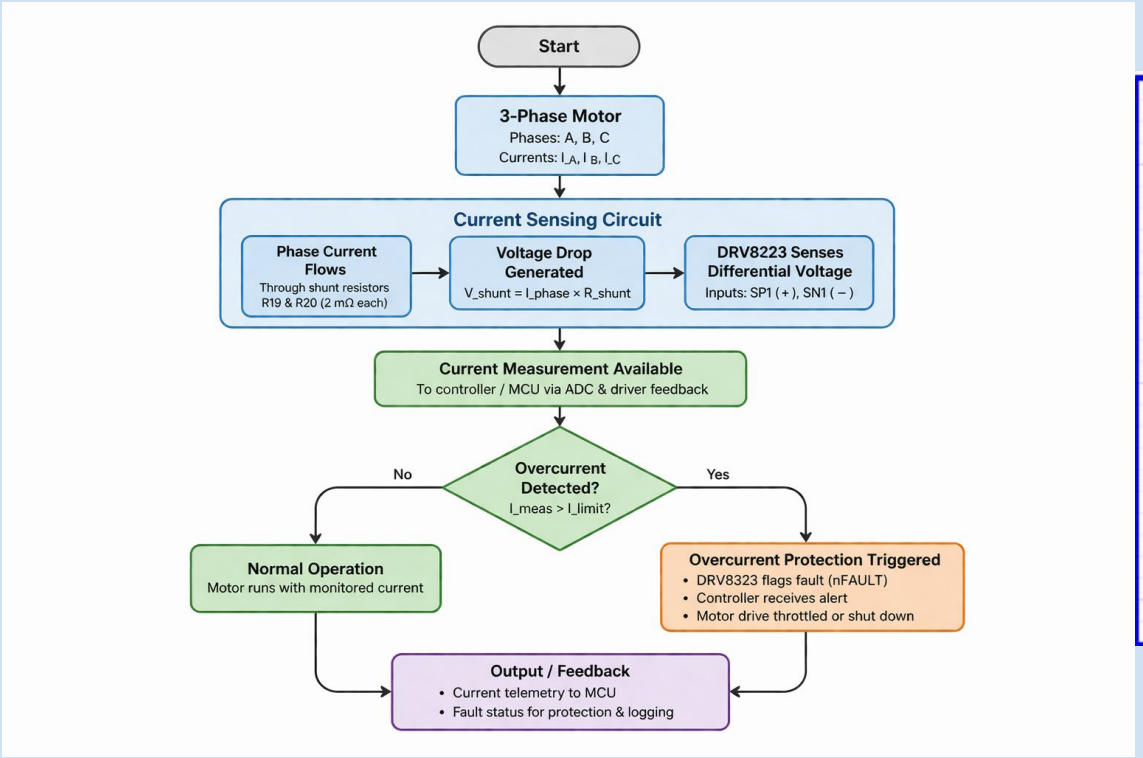


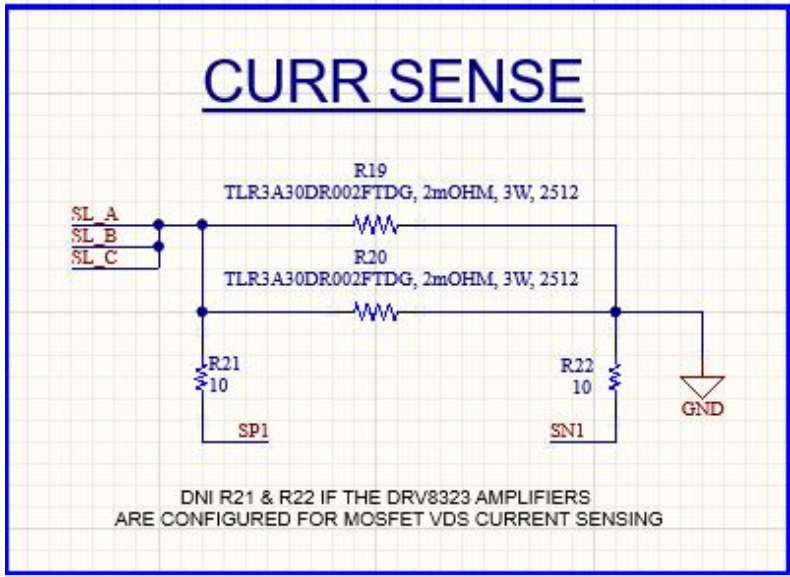


ESC Board / PCB Layout

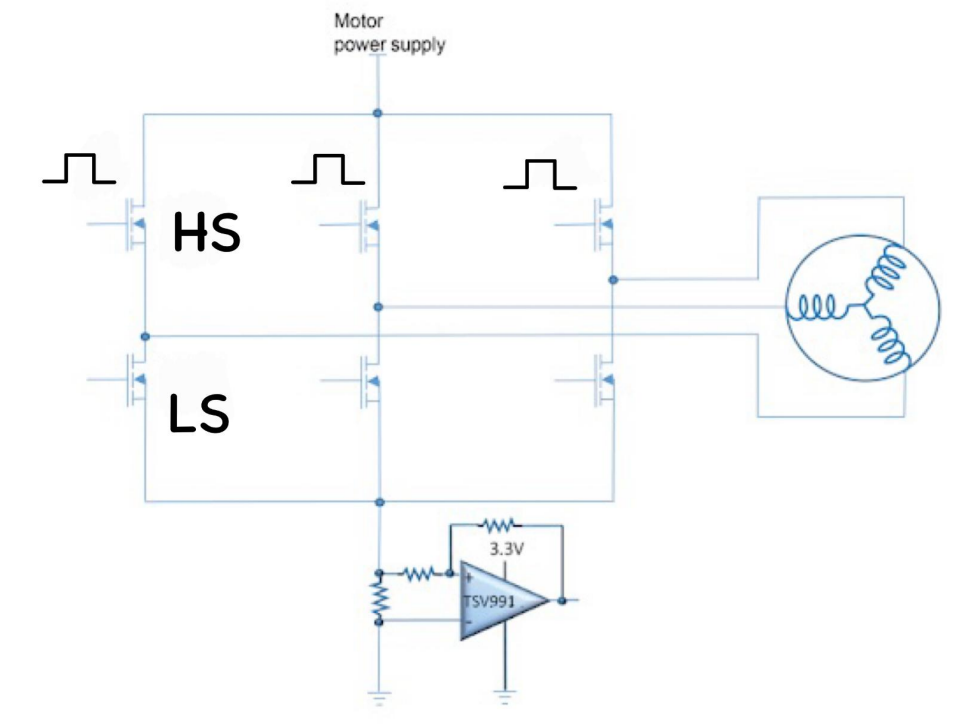


ESC PCB - Block Diagram



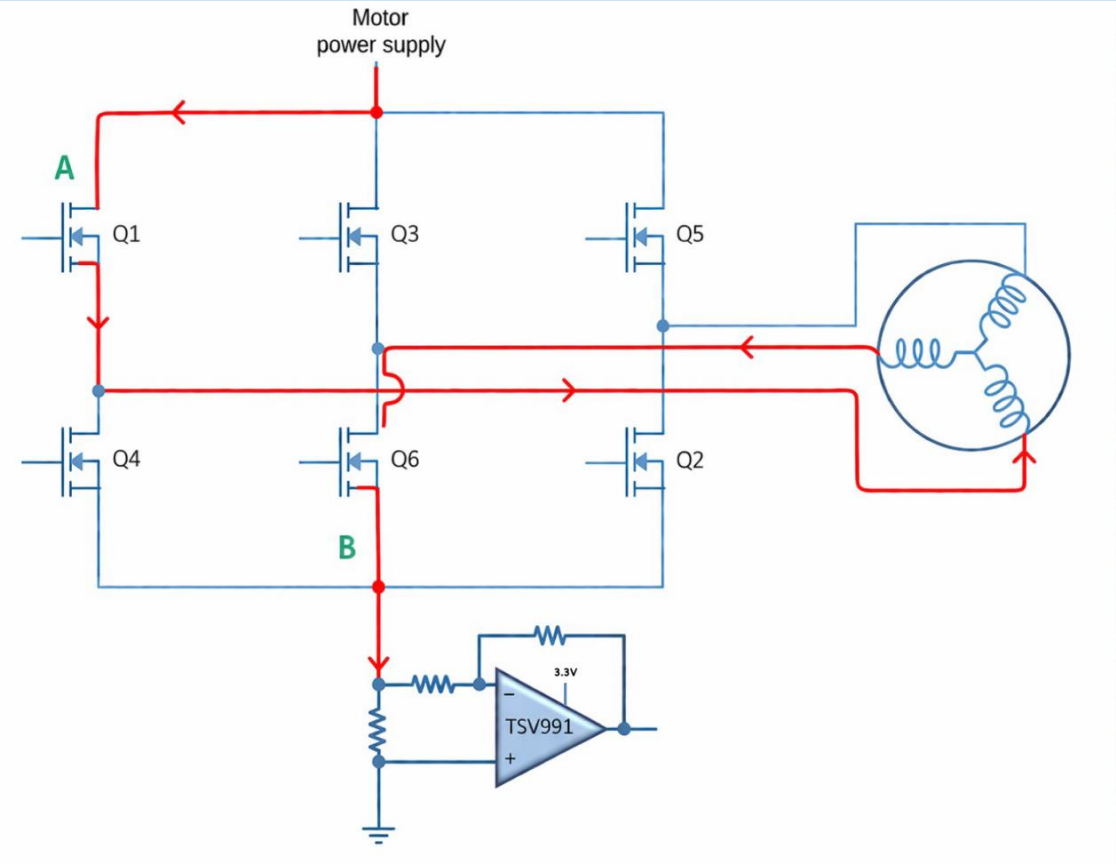


ESC PCB - High Side & Low Side MOSFETs



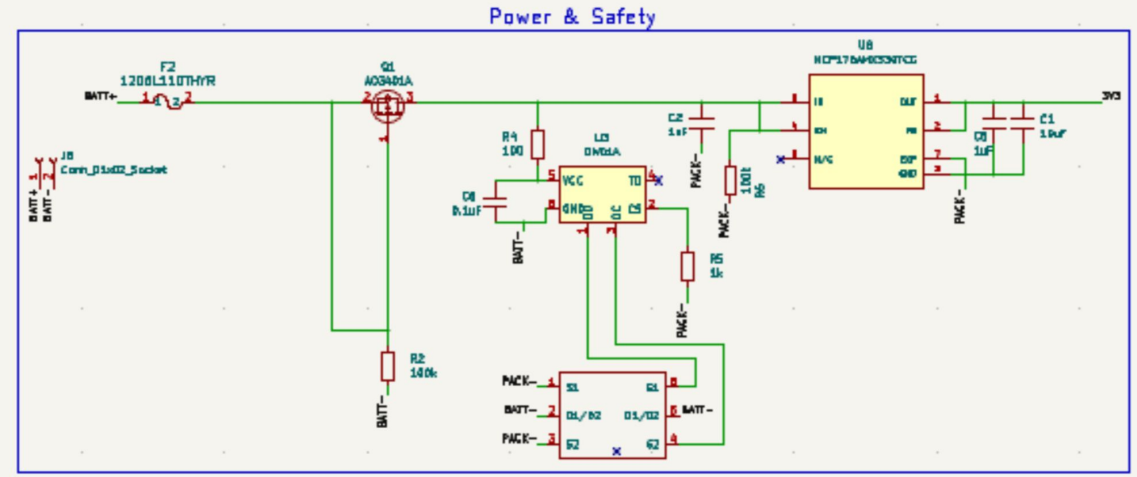
1. BLDC motor control using 3 phase PWM
2. Allows precise speed and torque control
3. Supports commutation of the three motor phases
4. Can support closed-loop control with MCU feedback
5. Allows precise speed control

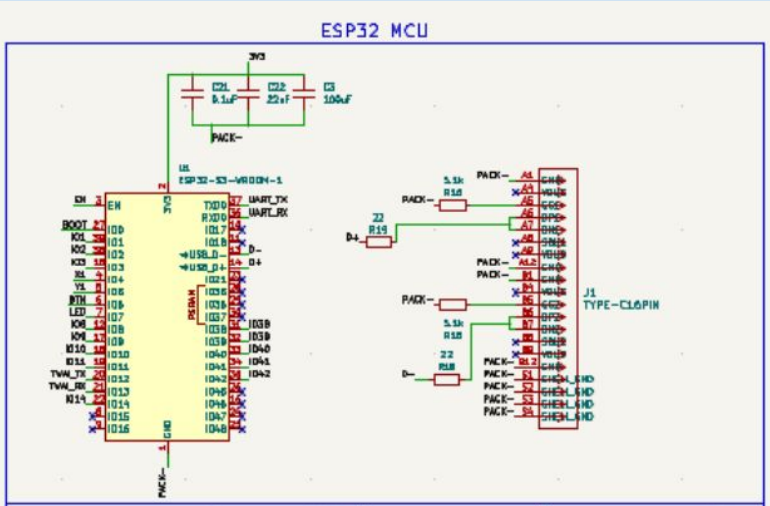
ESC PCB - High Side & Low Side MOSFETs

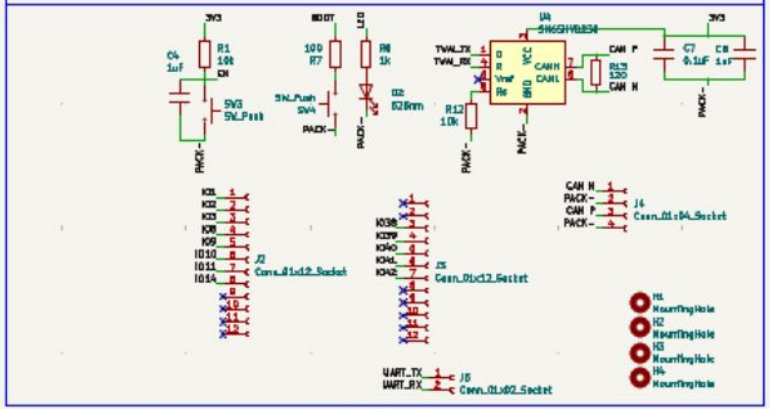


1. Current travels from the supply through the high-side FET of phase A
2. It then flows through the motor windings toward phase B
3. The low-side FET of phase B provides the return path to ground
4. Switching node voltage is scaled by a voltage divider
5. Comparator checks the divided voltage against the 3.3V reference
6. There is 6 sequences of HS and LS; in the correct order, the motor will rotate.

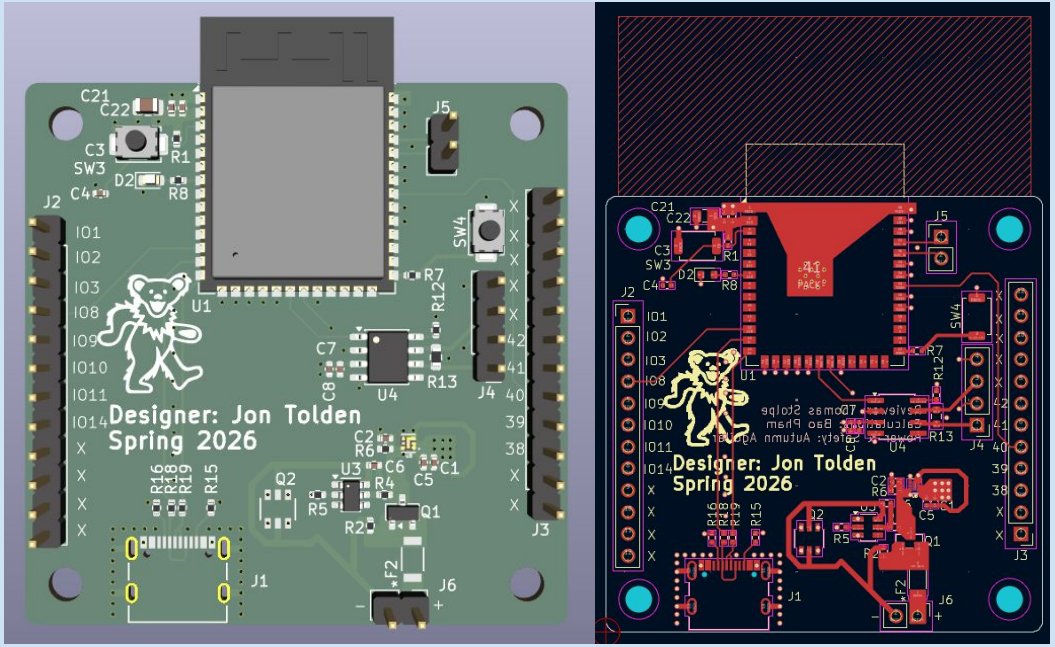
RF Receiver & Sensor Interface Control



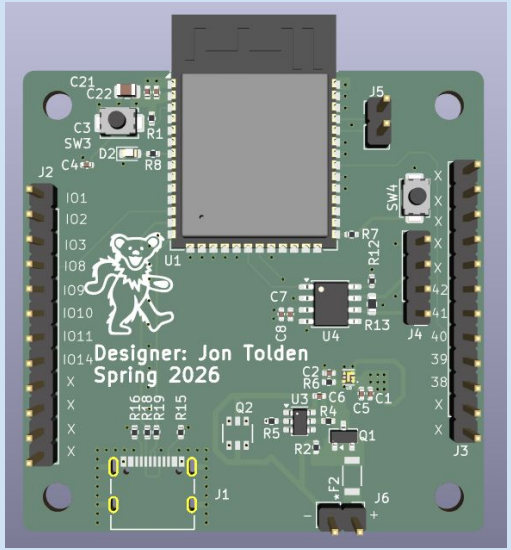




RF Receiver & Sensor Interface Control (Cont.)

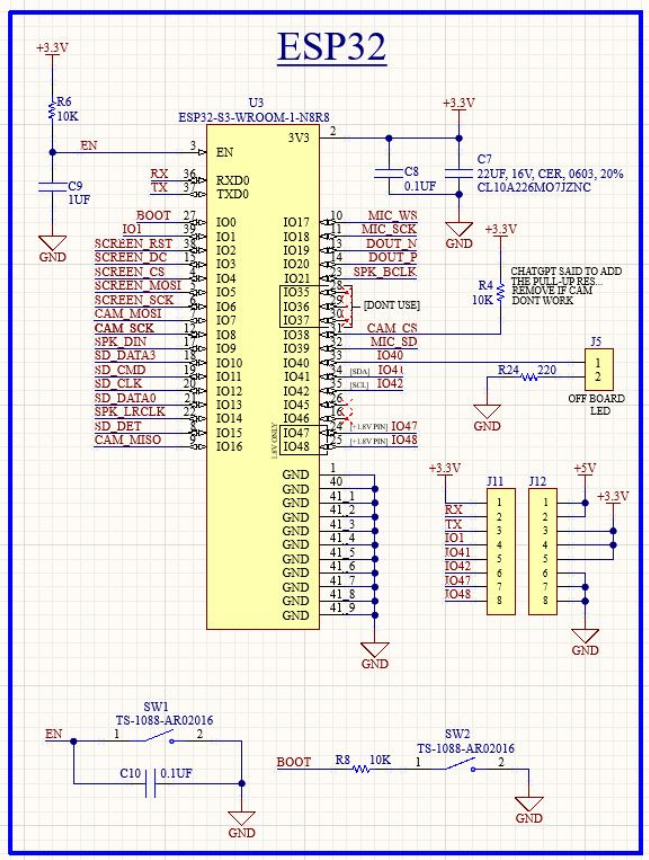


RF Receiver & Sensor Interface Control (Cont.)

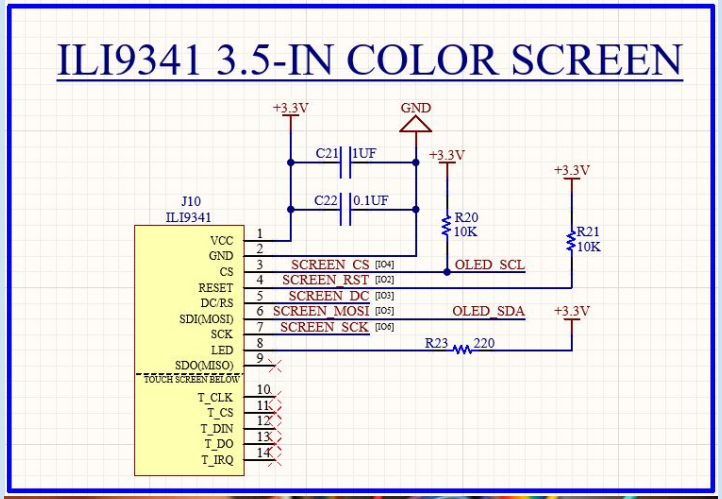


- Receives commands from remote
- Passes commands to each STM32 through CAN bus
- Receives motor current, board temperatures, and RPM from STM32s
- Battery level detection

RF Remote PCB



- The remote operates on an ESP32-S3 MCU
- The ESP32 was chosen for its beginner friendly environment and bluetooth capabilities
- 14 to 15 weeks to design and test PCB RF circuitry is ambitious



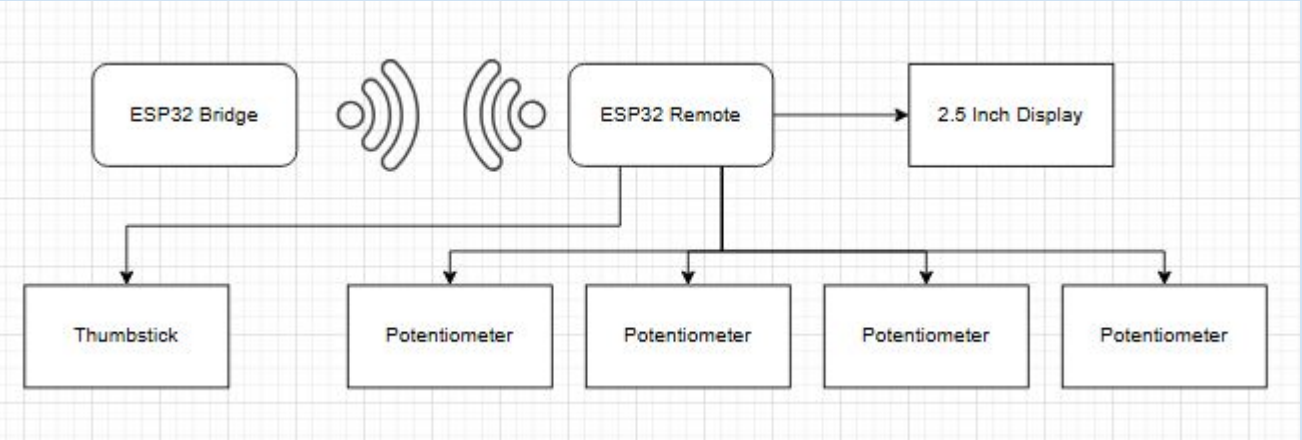


RF Remote PCB

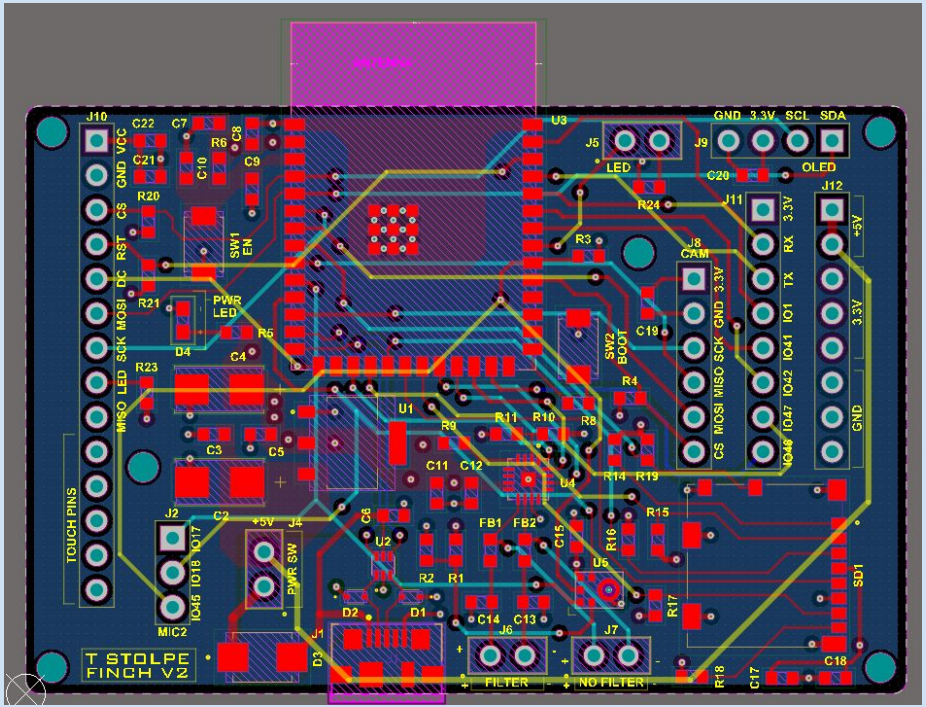
- The remote features a screen that will be able to display any self-diagnosed problems the system has
- Over-Current
- Fault codes with BLDC driver chip\

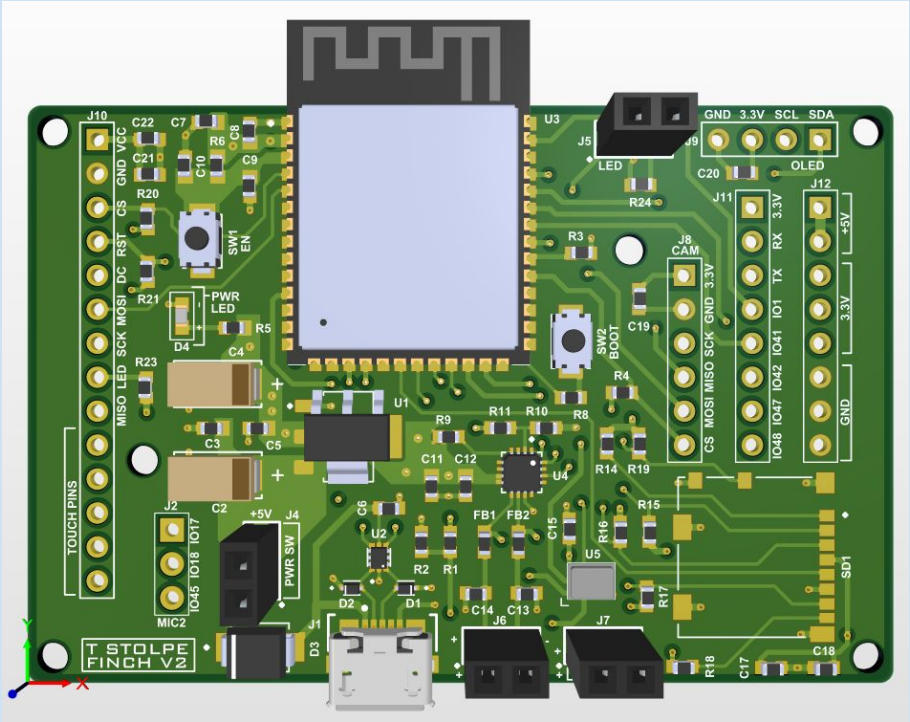
- Will also display telemetry for each motor.
- RPM , Motor Current , PCB temperature of each PCB , and Fault codes from the BLDC driver chip

RF Remote PCB

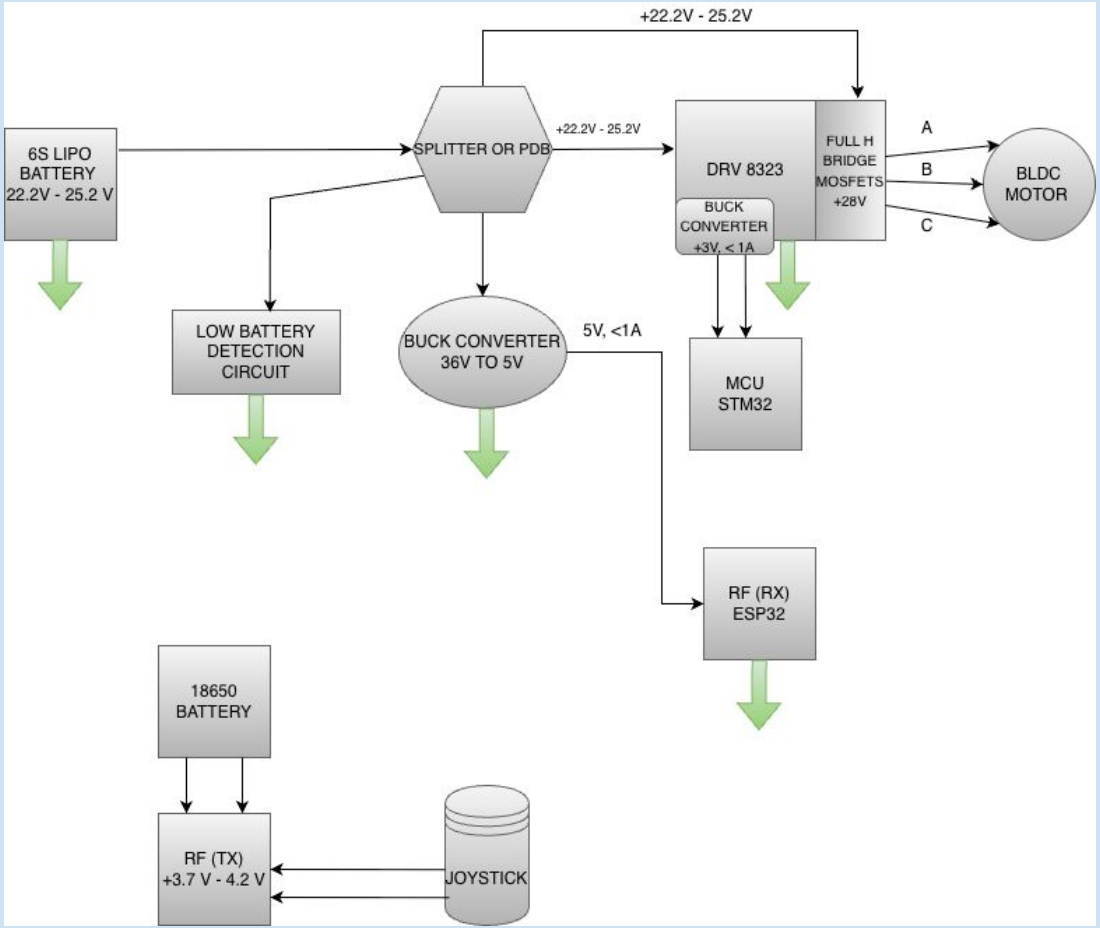


RF Remote PCB

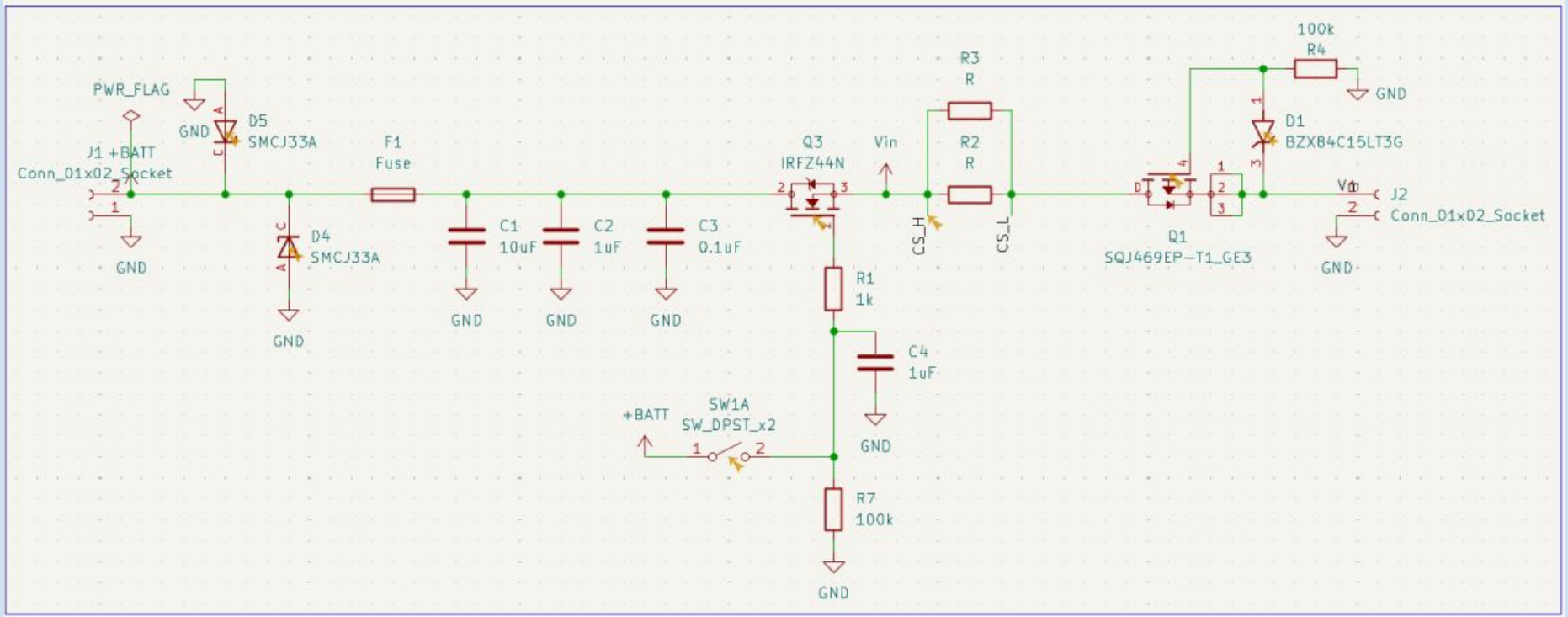




Power Block Diagram

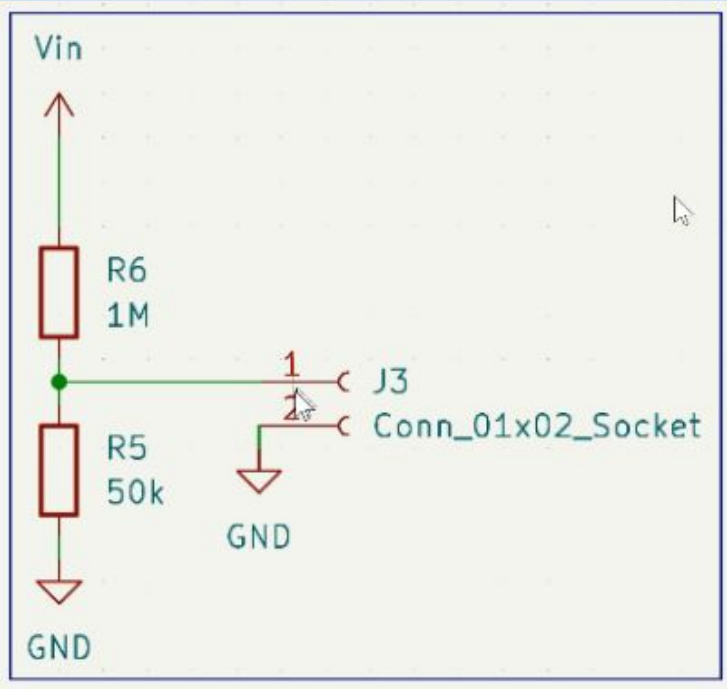


Power Circuit Schematic



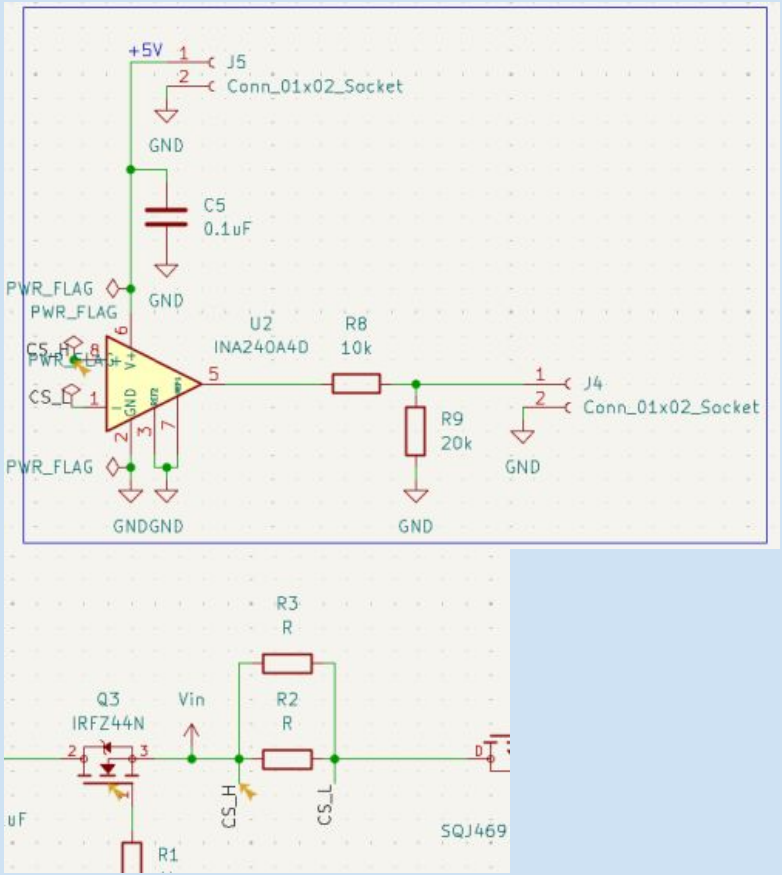
- A switch is connected to the gate of a MOSFET to allow for a high current power switch
- CS_H and CS_L go to a current sense amplifier to measure the total circuits current. Too high = automatic shutdown
- Q1, R4, and D1 form a reverse polarity circuit. A 6S Lipo battery will turn a circuit into plasma very quickly if plugged in the wrong way.

Power Circuit Schematic



- A voltage divider is formed at the battery input with a 1M and 50k resistor.
- The middle of the divider goes to an ADC pin
- As the battery drains, the analog value on this pin decreases, allowing for battery level detection
- At 70V (large voltage spikes from a very strong battery), the voltage going to the ADC pin becomes 3.33V, which is nearly the max voltage the MCU pin can handle
- A zener diode was later added to clamp the ADC pin to 3.5V with a 2.2k current limit resistor in series with the zener
- As the battery voltage approaches its unsafe under-voltage level, the system prevents the motors from turning on, to prevent the battery from getting damaged

Power Circuit Schematic

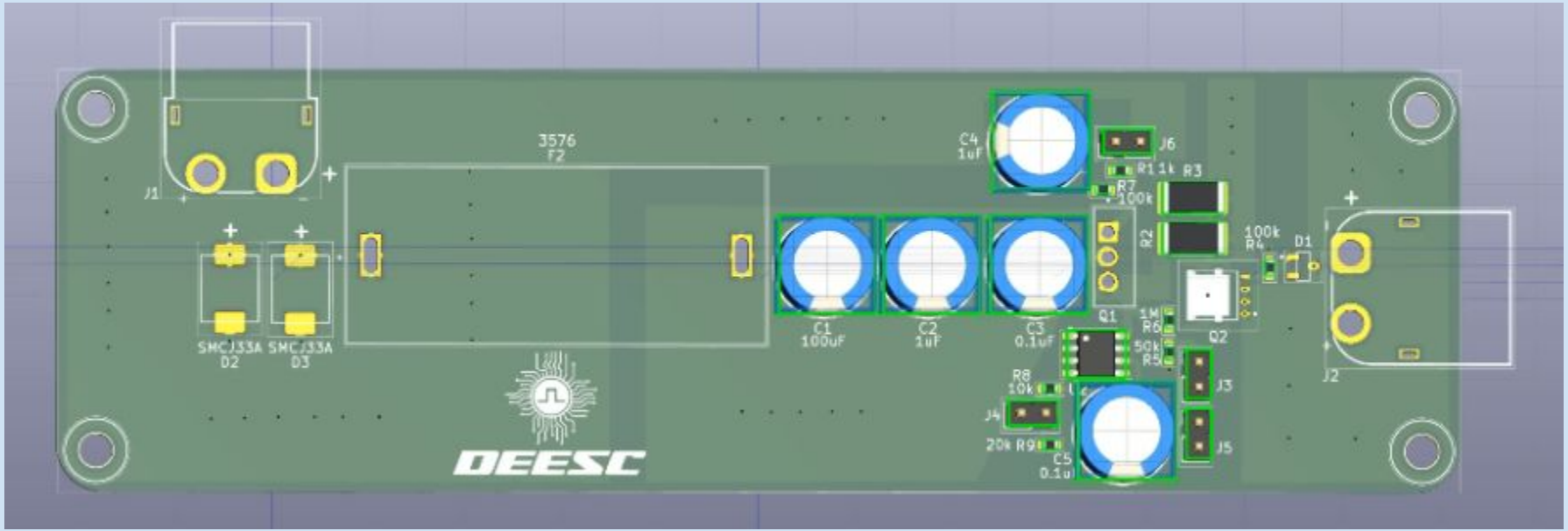


- U2 is the current sense amplifier measuring a very small voltage drop across 2 milliOhm resistors, R2 and R3.
- This voltage difference is amplified with a gain of 200

- U2's Vout = Gain * (IN+ - IN-)
- This output rails to the 5V V+ supply of the amp when the voltage drop across R2 and R3 exceeds 25mV (5V / 200 gain). This

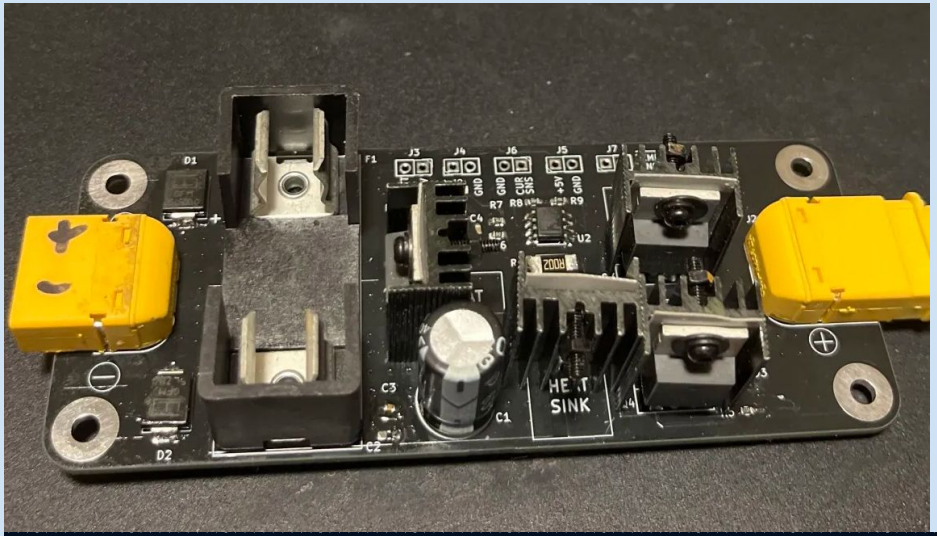
- The output of U2 goes to an ADC pin on the Diagnostic ESP32 which can be programmed to make decisions on what to do at certain current levels

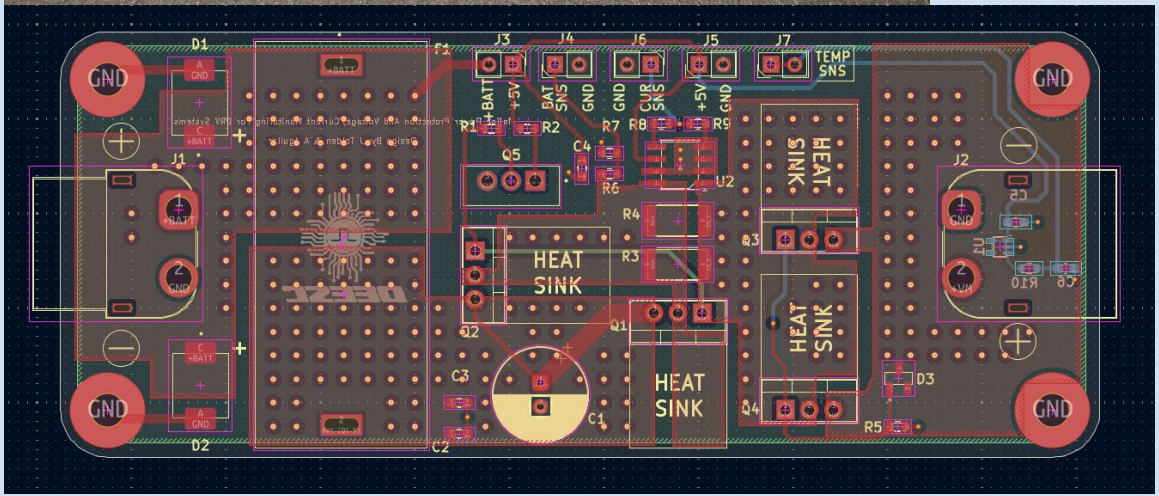
Power Circuit Schematic



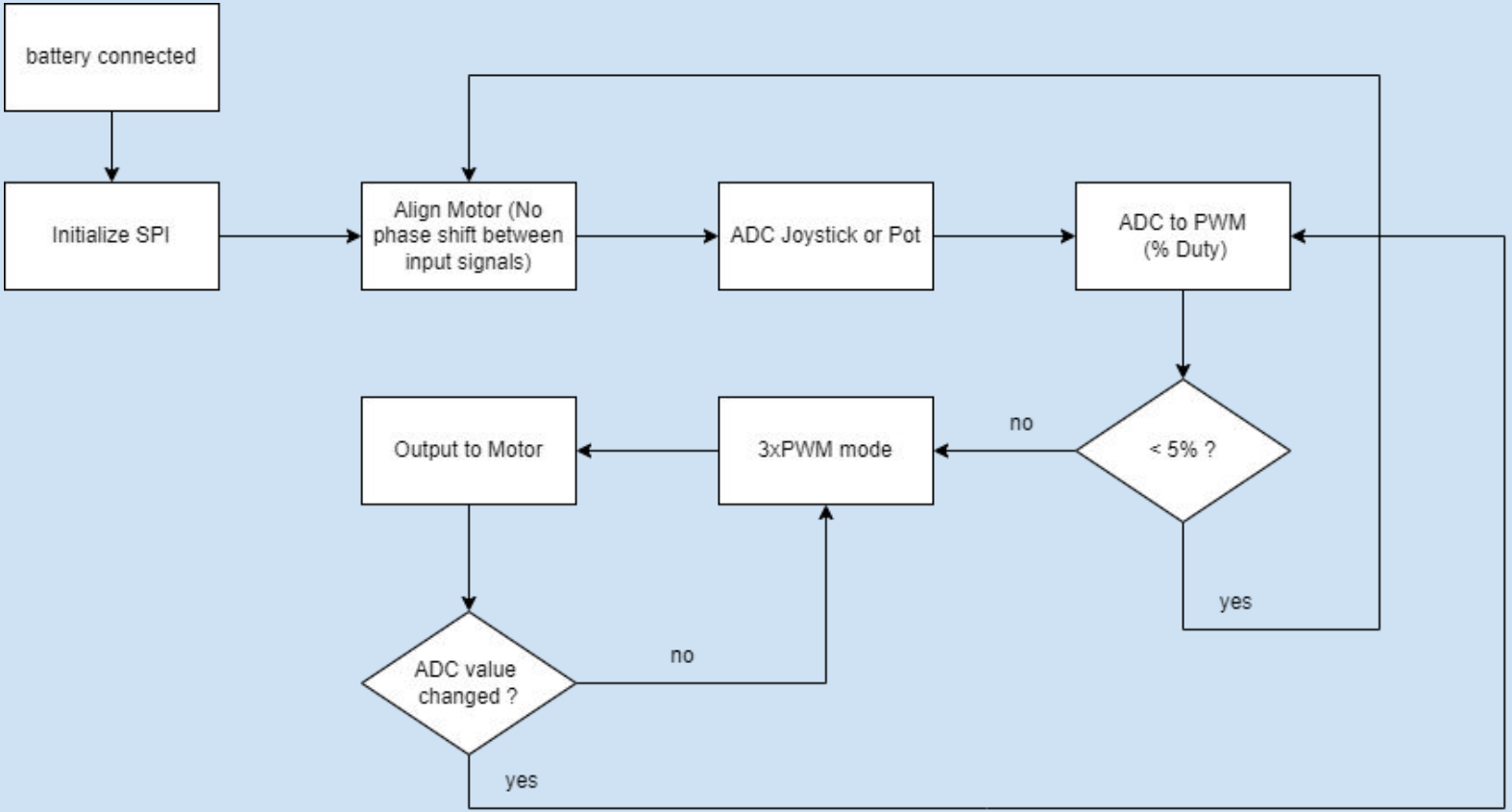
- Polygon pours across 3 layers carry high current with vias to manage heat to some degree.
- A large fuse (10mm x 38mm) with a 30A current rating is placed at the battery input. A 4S Lipo battery can output 200+ Amps during a short.

Power Circuit Layout

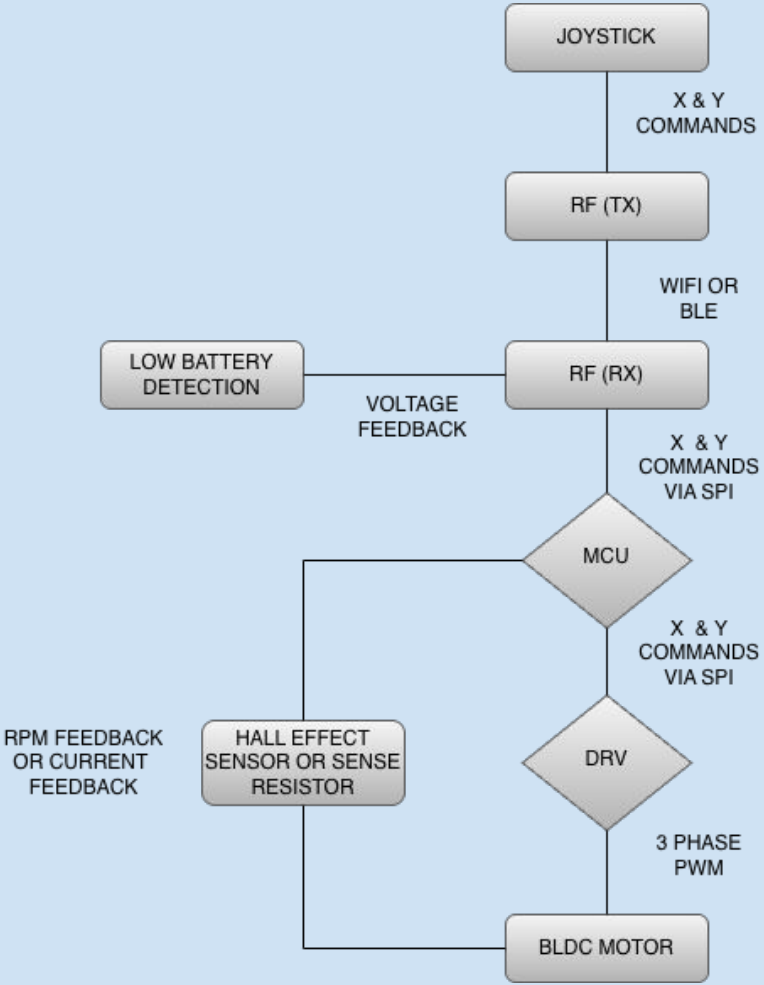




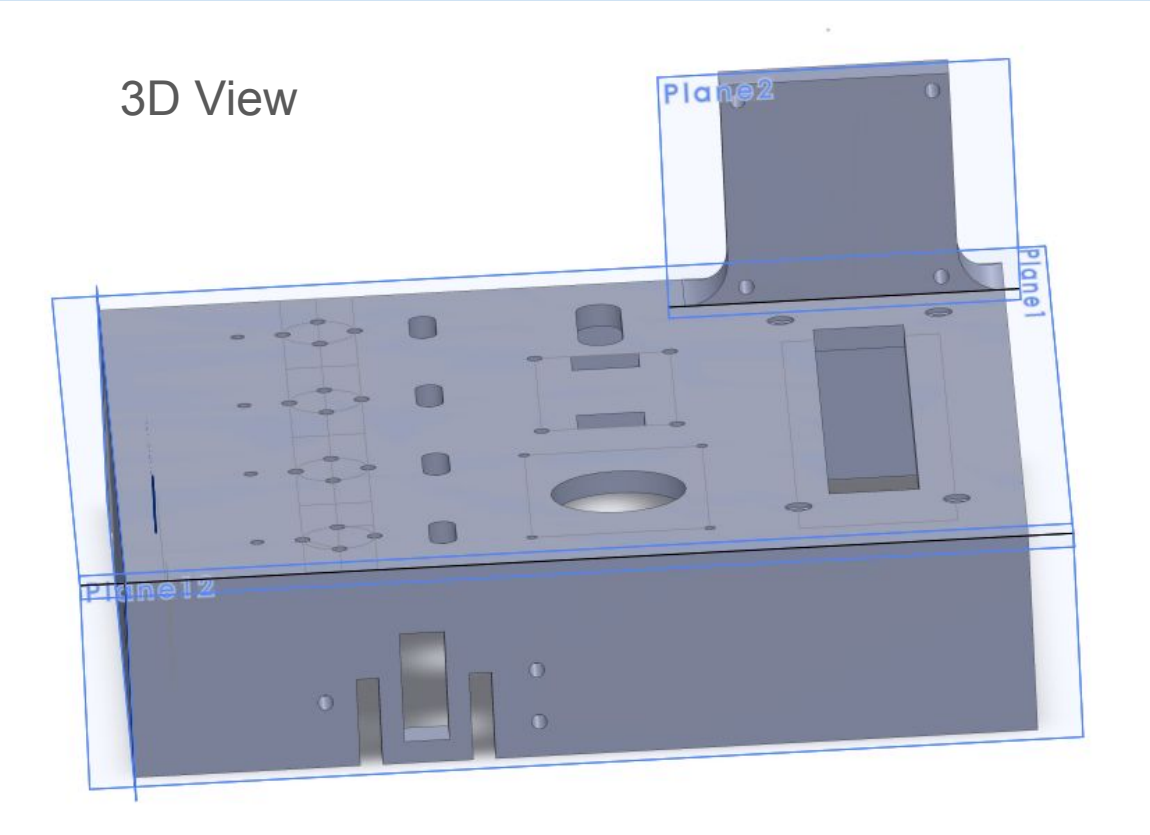
Software Design (flowcharts and results)



Signals Block Diagram



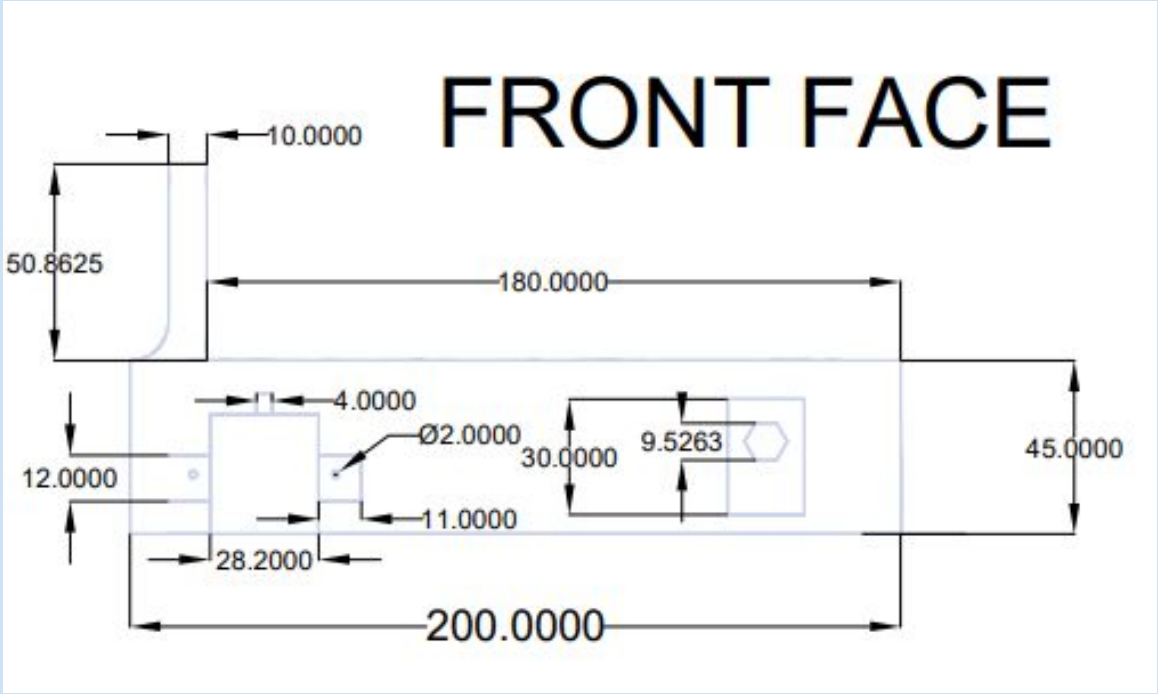
Mechanical Design (sketches/models and results)



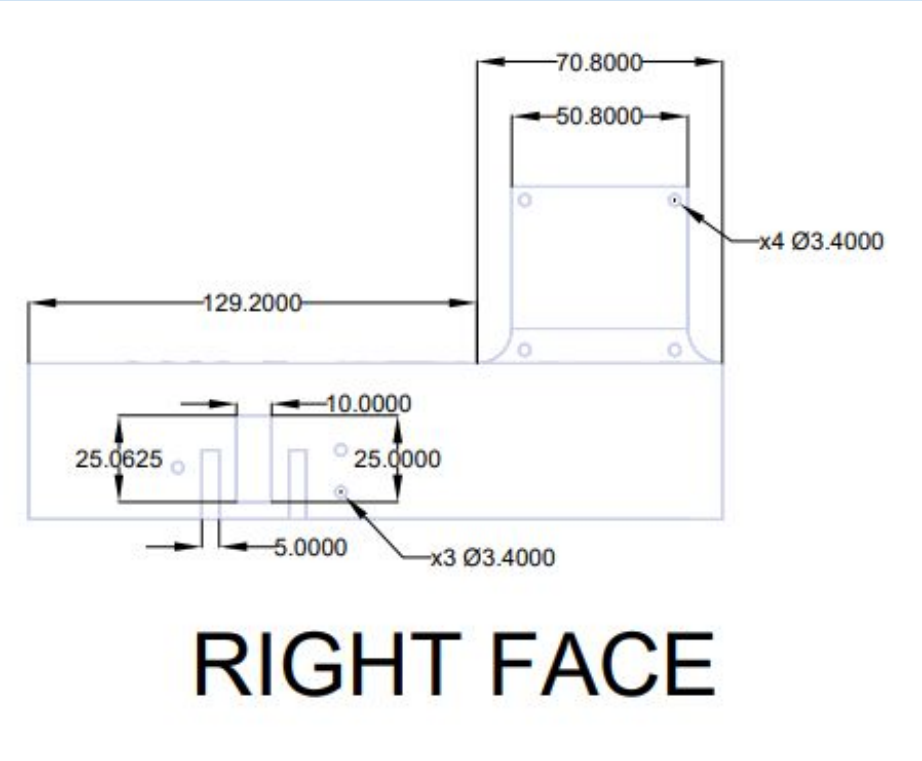
The test supports:

- 4 BLDC Motors
- 11 PCBs
- 4S or 6S Lipo Battery
- Potentiometer

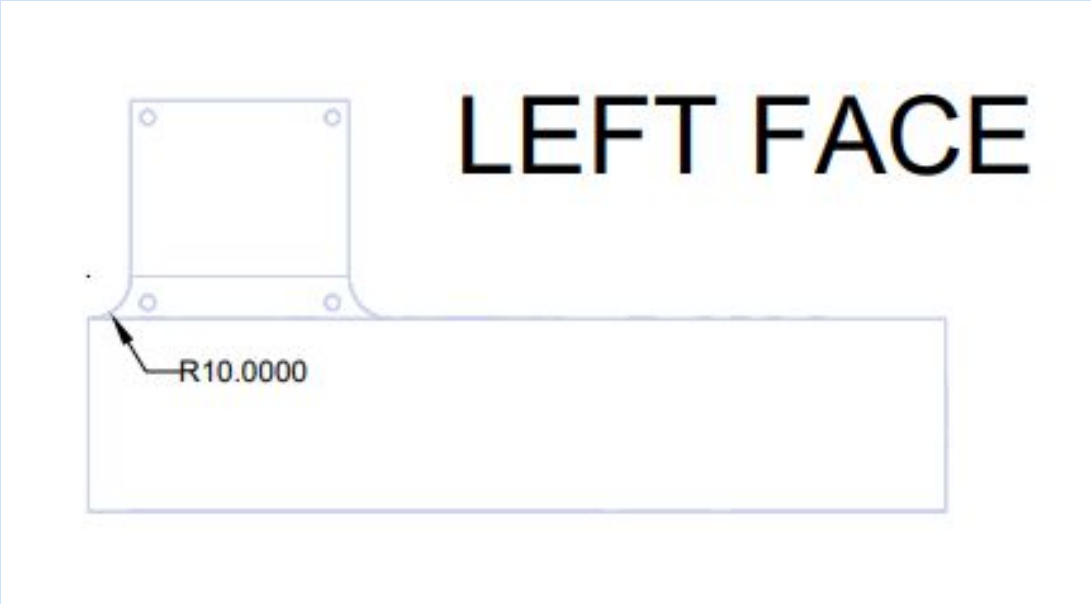
Mechanical Design (sketches/models and results)



Mechanical Design (sketches/models and results)



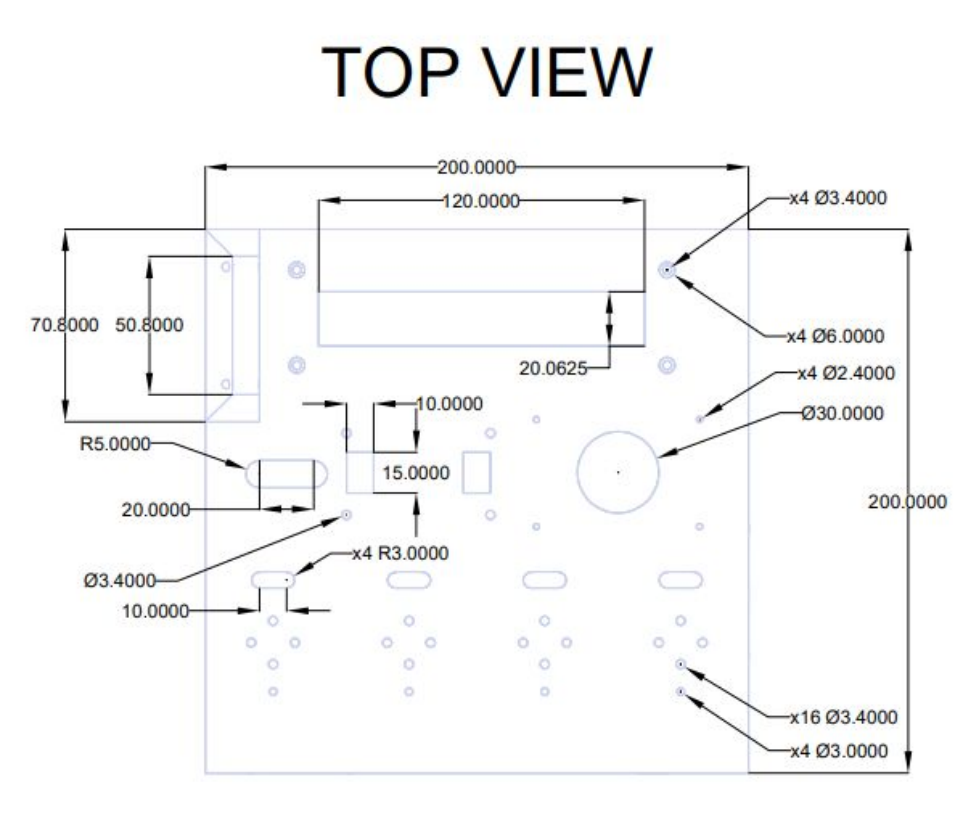
Mechanical Design (sketches/models and results)



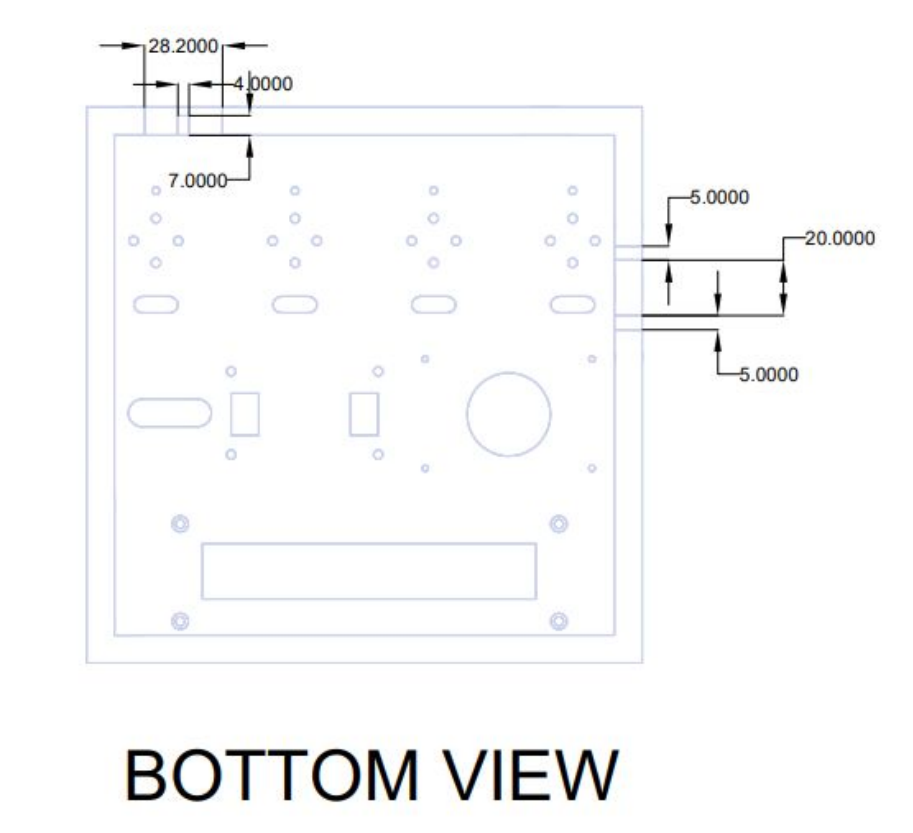
Mechanical Design (sketches/models and results)



Mechanical Design (sketches/models and results)



Mechanical Design (sketches/models and results)



Resource Reports

Cost

- PCB: $250
- Power stage components: $50
- Control electronics: $75
- Connectors and wiring: $13

Power/Electrical Specs

- Input voltage range: 5-30 V
- Logic rail outputs: 5V and 3.3 V
- Current target: up to 20 A
- Low battery detection threshold: 14 V

Build Status

- Power stage PCB selected and assembled
- Control circuitry integrated with power stage
- Wiring and connectors installed for testing
- Enclosure and screen housing in progress

Problems & Solutions

1. The RF board USB inputs were unintentionally tied together, causing input conflicts.

Solution: The USB connections were corrected and separated.

2. An NMOS device was mounted in the wrong orientation.

Solution: The transistor was reinstalled in the correct direction.

3. A manufacturing error resulted in a 0 Ω resistor, which shorted the circuit and disrupted CAN bus operation.

Solution: The incorrect part was removed and replaced by soldering in the correct resistor value.

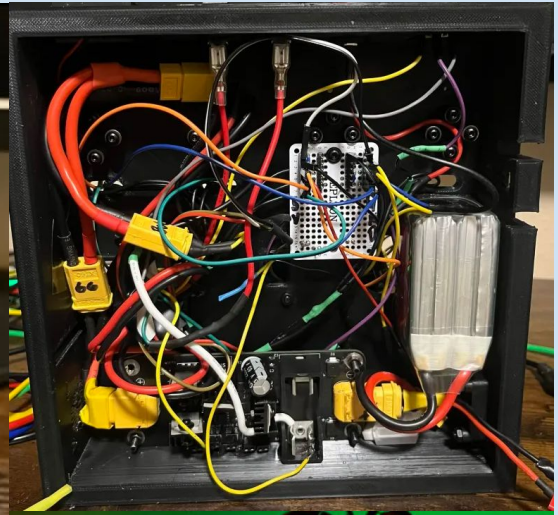
4. During testing, a driver wire was pulled loose and the ground pin was damaged.

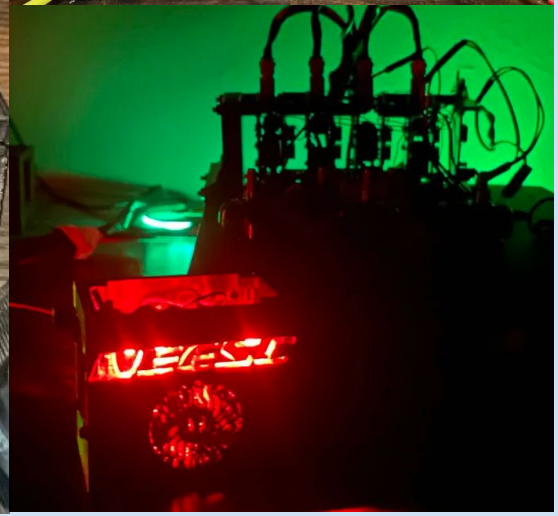
Solution: The wire and ground connection were repaired and resoldered successfully.

Remaining Work

Task	Owner
Finalize 3D model/mounting stand	All team members
Finish ISENA unlock and calibration pin logic	Autumn & Bao
Fully implement CAN bus and multi motor support	Bao
Finish ESP-32 features and Bluetooth telemetry	Jon
Complete battery input and protection circuitry	Autumn
Finalize bidirectional motor control	All team members
Calibrate temperature sensing	Autumn & Jon
Clean up software for unused pins	Bao & Jon
Verify all requirements are meant	All team members


HDMI











QUESTIONS OR COMMENTS?